

4. Simple Linear Regression III

MATH 261A: Regression Theory and Methods

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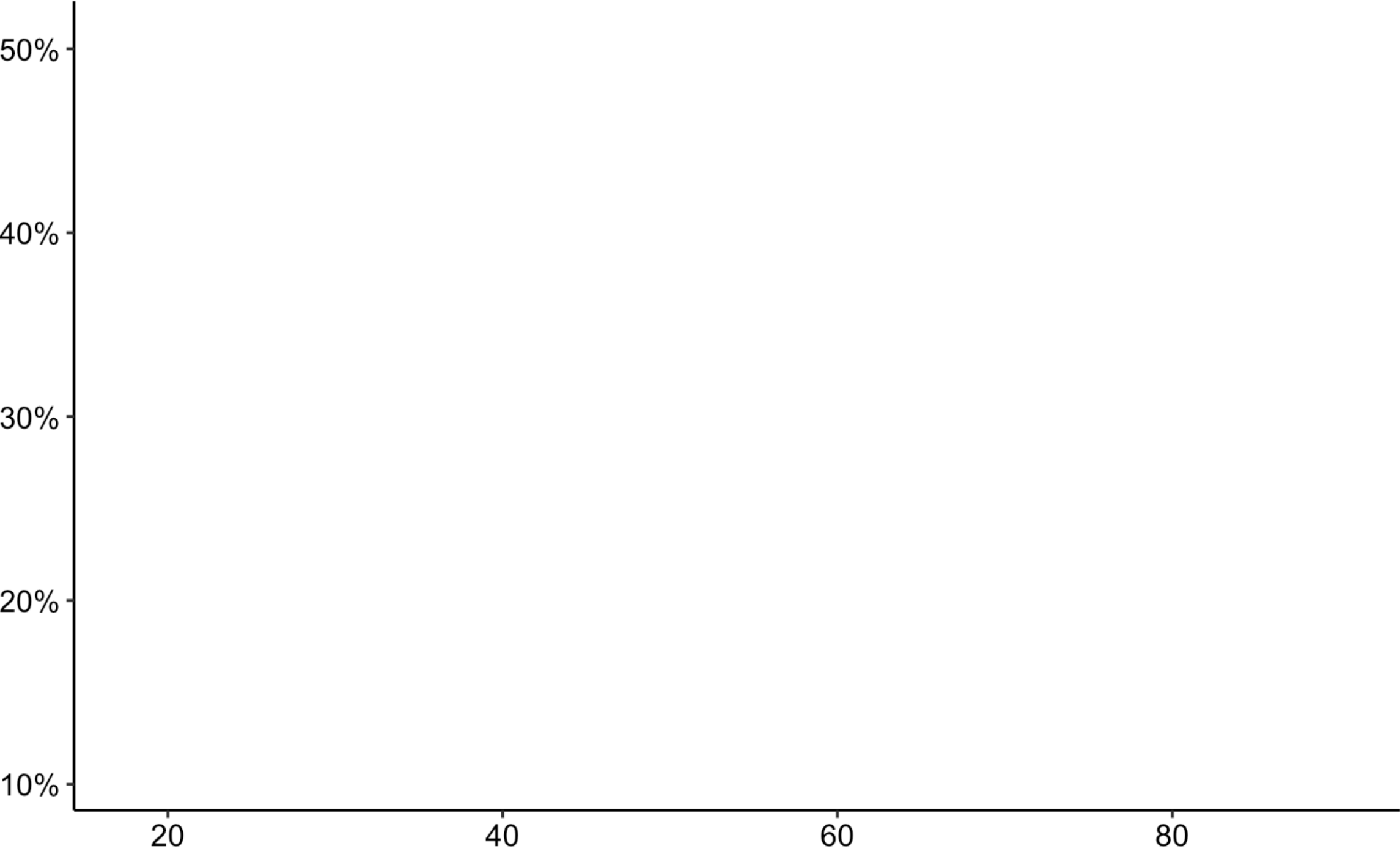
Coding exercises

Use simulation to demonstrate that for the simple linear regression model, even if the errors have unequal variances, as long as they have mean zero, the least squares estimators are unbiased.

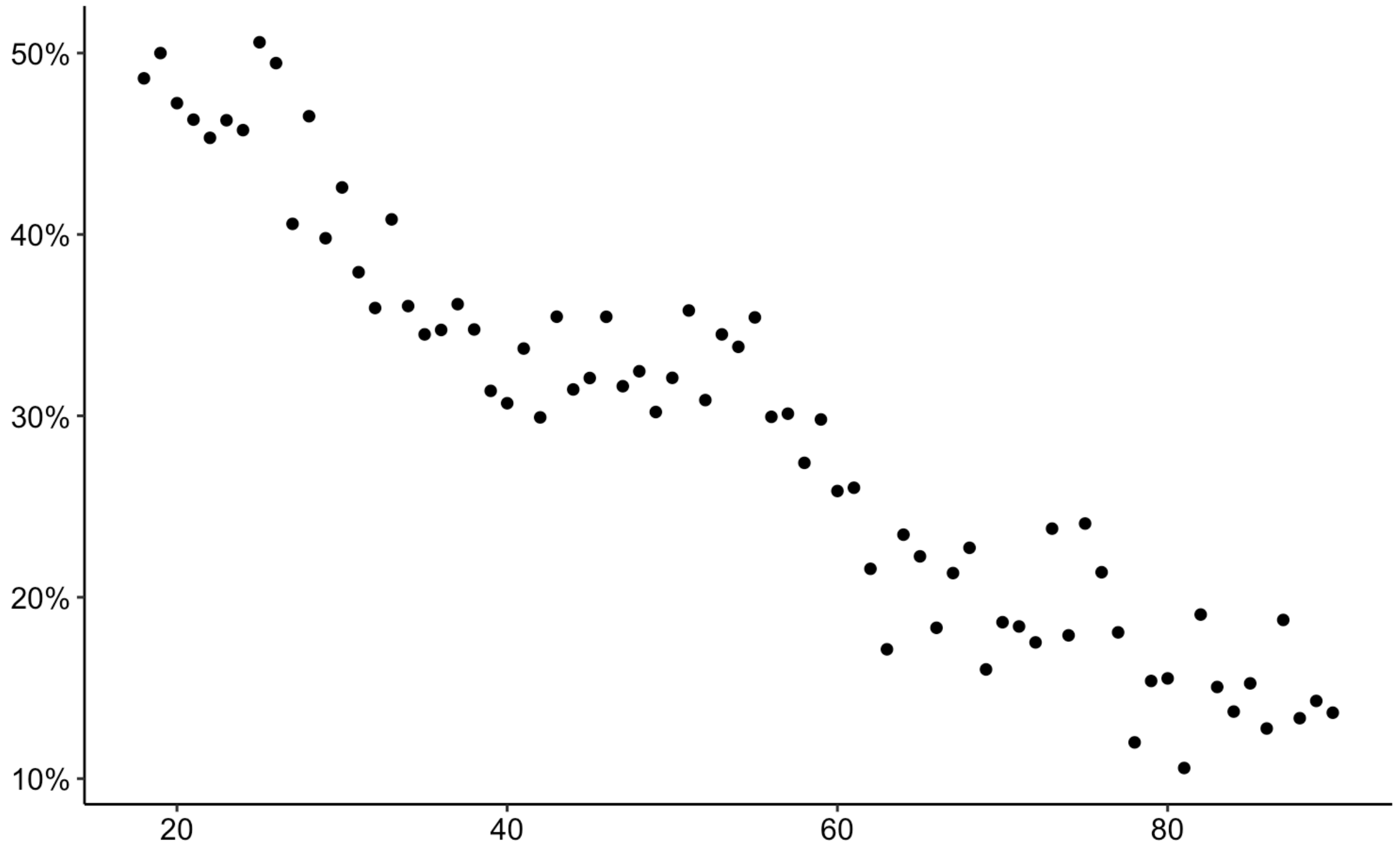
Class outline

- Understand simple linear regression from the analysis of variance point of view
- Discuss F tests
- Coefficient of determination (r -squared) and correlation (r)

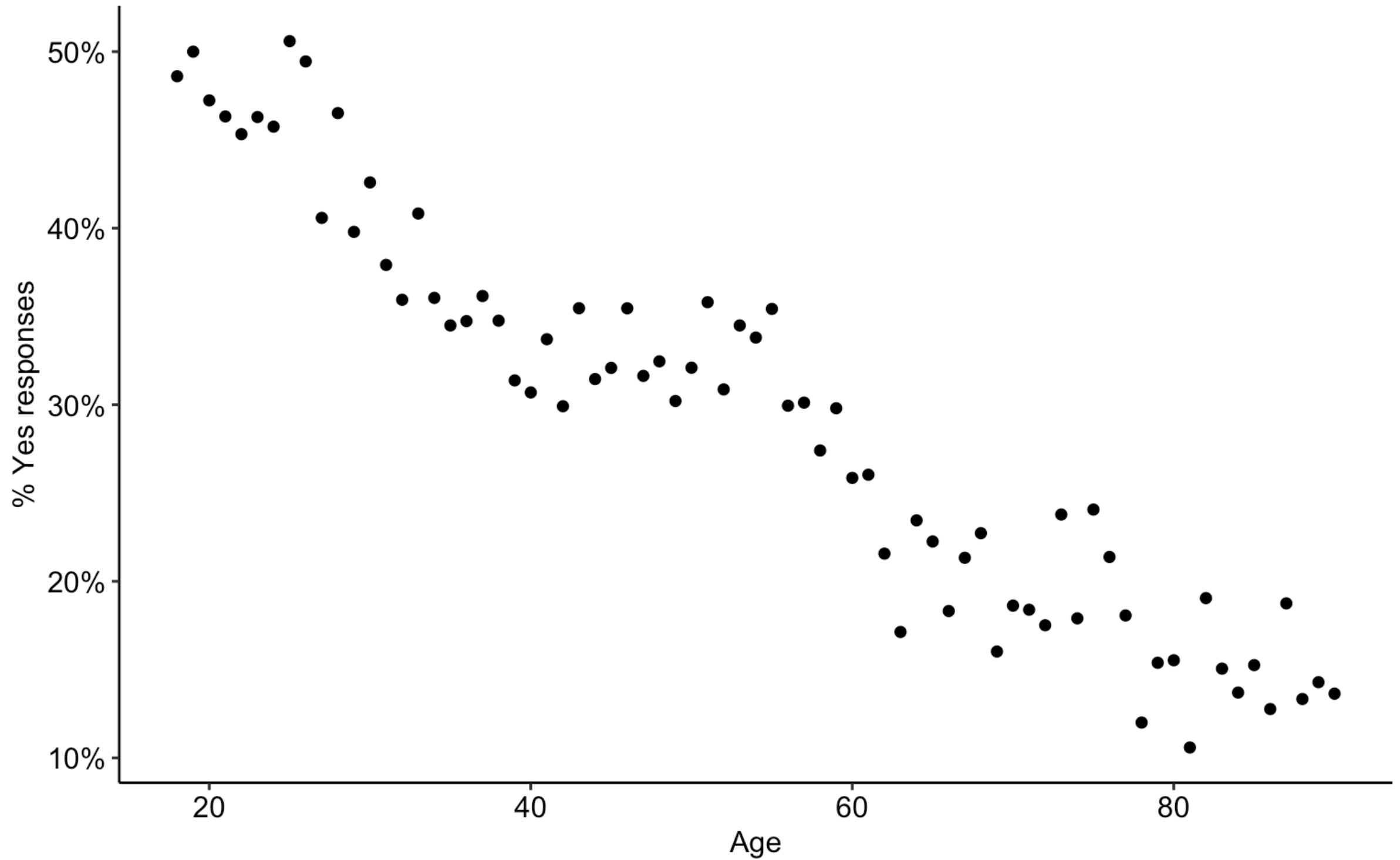
Warm-up: Scatterplot charades



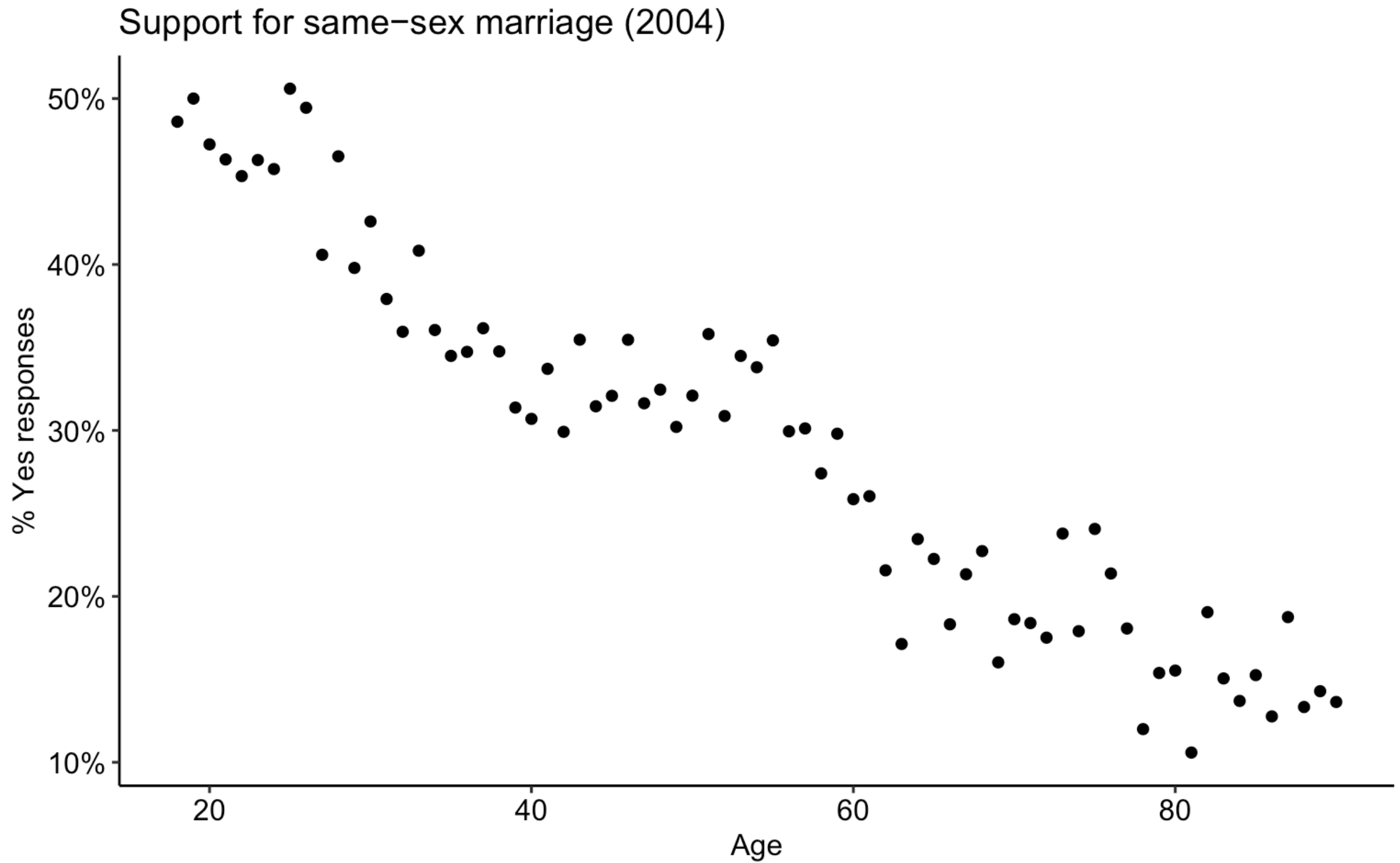
Warm-up: Scatterplot charades



Warm-up: Scatterplot charades



Warm-up: Scatterplot charades



History of ANOVA

Broadly speaking, analysis of variance (ANOVA) assesses differences between group means by comparing the variation between group means with the variation within each group.

In his book **Statistical Methods for Research Workers** (1925), R.A. Fisher developed ANOVA methods for assessing whether differences in wheat crop yields between groups receiving different fertilizers were statistically significant.

Analysis of variance: intuition

ANOVA has historically been presented as a method for assessing the effect of a **categorical** predictor, but understanding its application to **continuous** predictors will help us develop a general framework for assessing the fit of linear models.

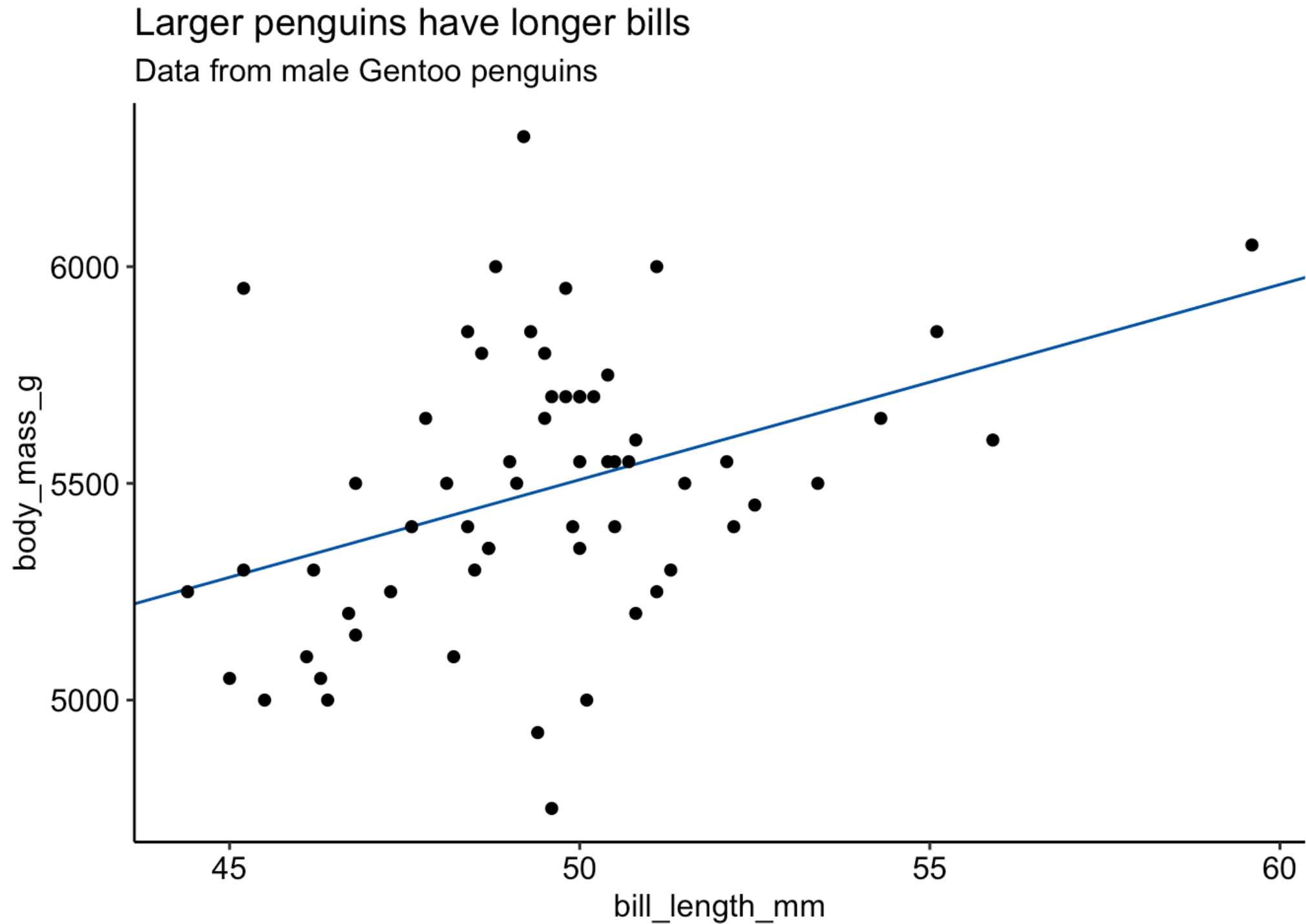
Intuition. The goal is to quantify and decompose the total variation in some outcome variable Y (ex. final exam score):

$$\begin{aligned} \text{total variation in } Y &= \text{variation explained by model} \\ &+ \text{residual variation} \end{aligned}$$

If the variation explained by the model is a substantial portion of the total variation, then perhaps the model is a good fit to the data.

Does this look like a good fit?

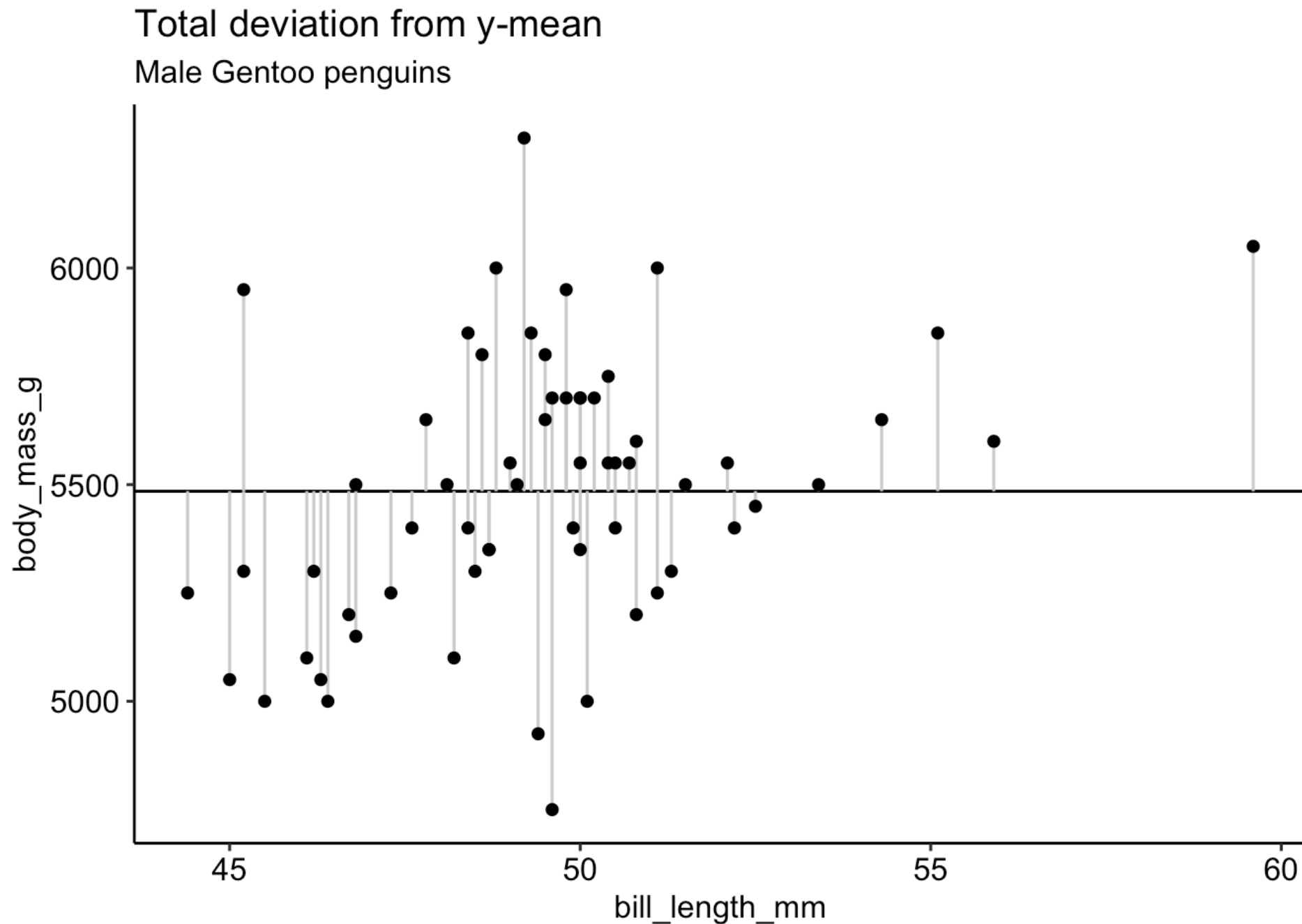
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Decomposition of total variation in Y

Consider the deviations from the mean, which represent the **total variation** in Y :

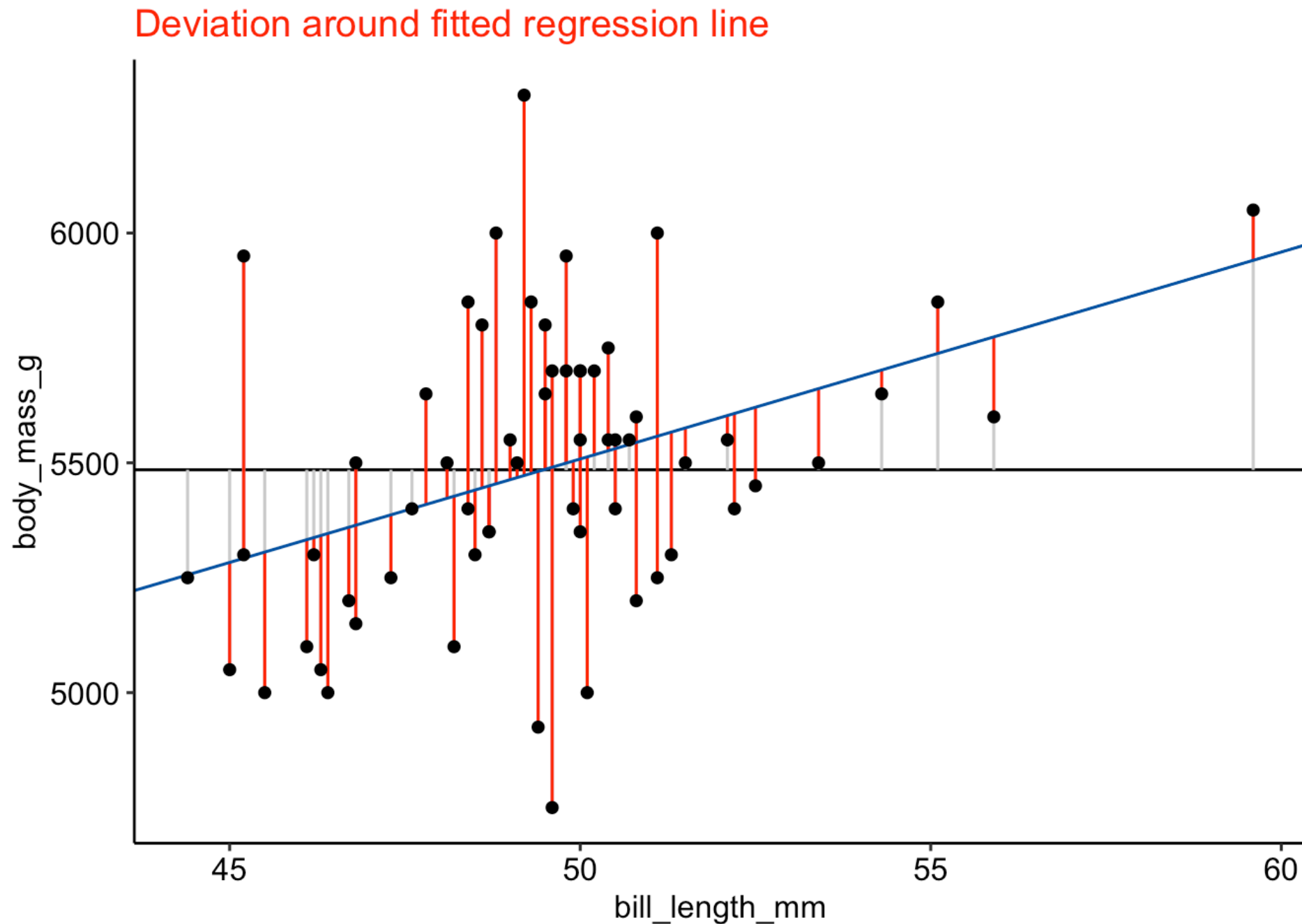
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Decomposition of total variation in Y

We can decompose these deviations into the **deviation around the regression line**:

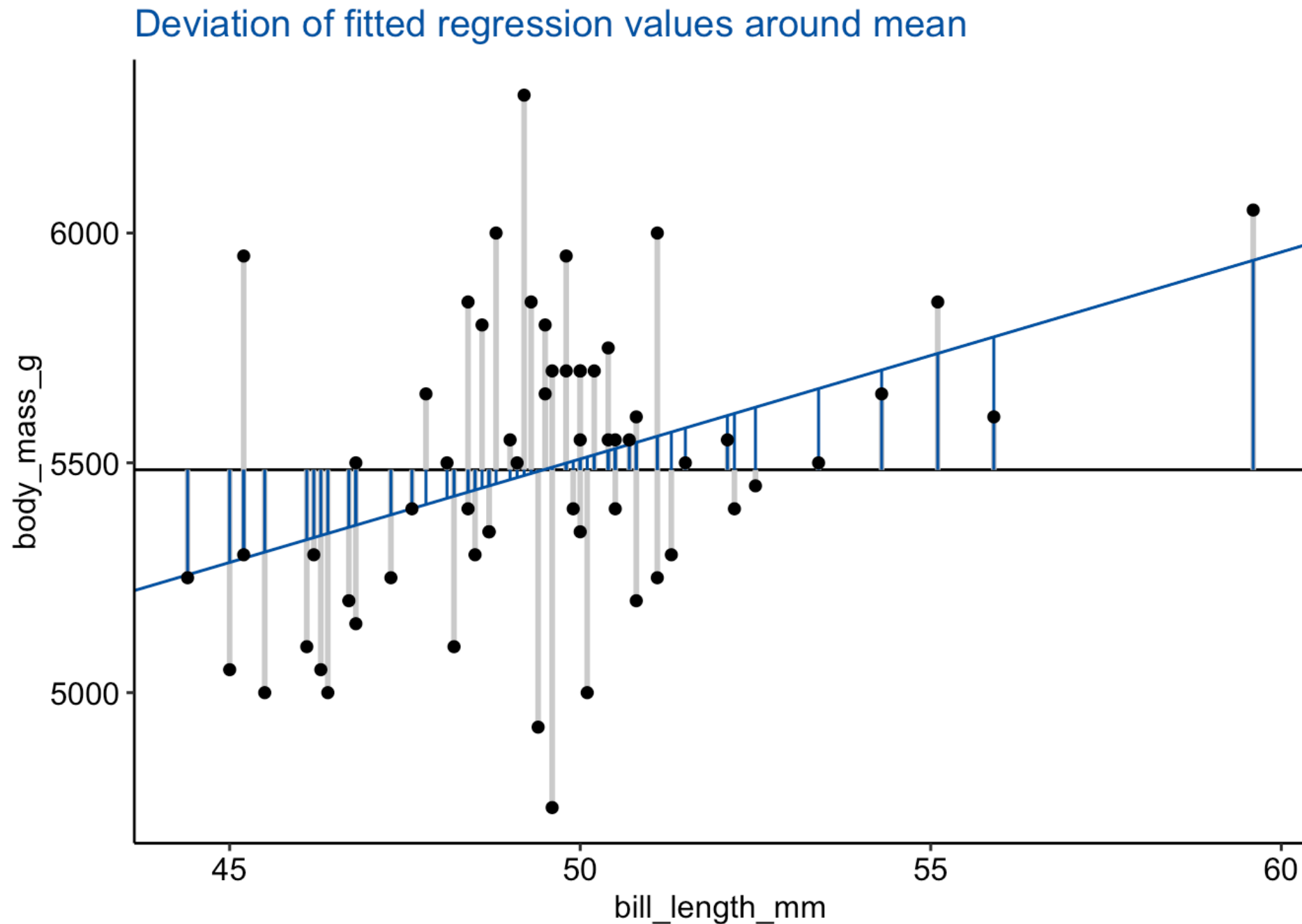
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Decomposition of total variation in Y

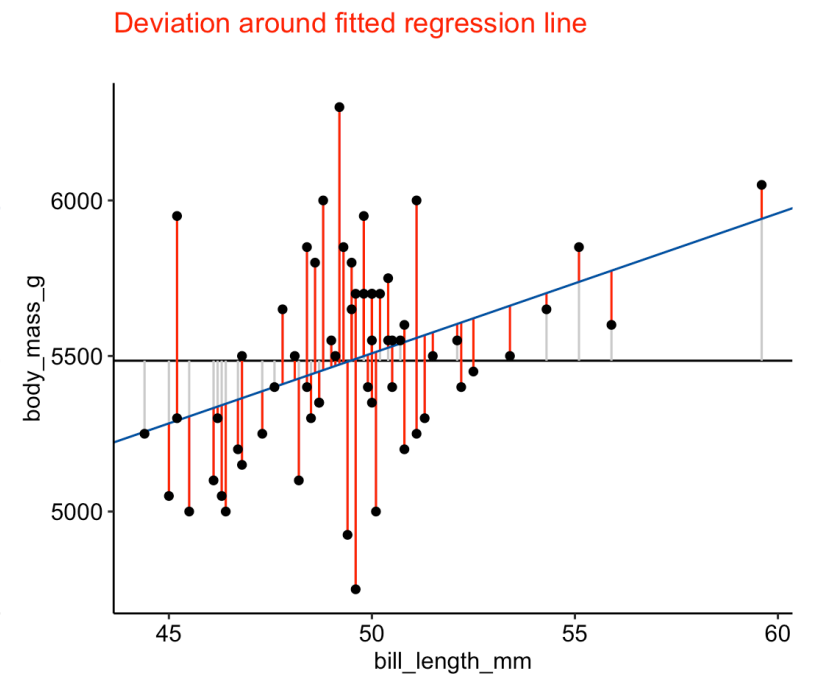
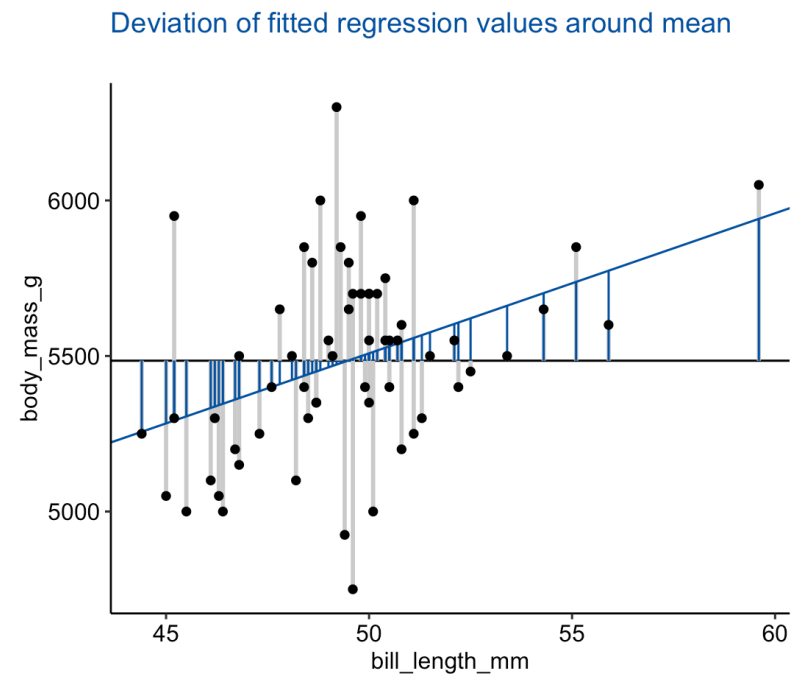
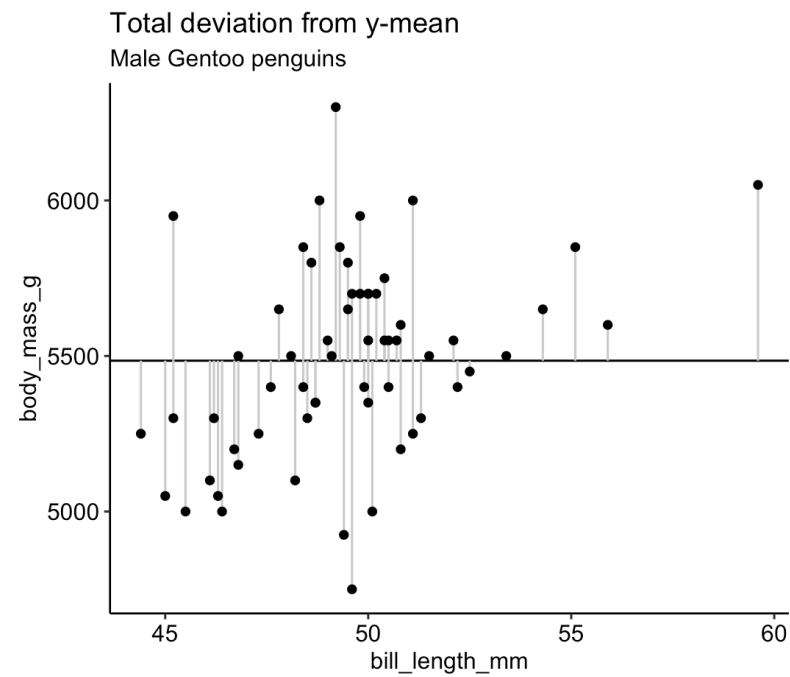
And the **deviations from the mean line to the regression line*:

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Decomposition of total variation in Y

► Code



Formal decomposition of total variation in Y

Let Y_i denote the i th response, and $\bar{Y}$ the sample mean of $Y_1, \dots, Y_n$. Let $\hat{Y}_i$ denote the fitted value for the i th unit based on the fitted regression line.

$$\underbrace{Y_i - \bar{Y}}_{\text{Total deviation}} = \underbrace{\hat{Y}_i - \bar{Y}}_{\text{deviation of fitted values around mean}} + \underbrace{Y_i - \hat{Y}_i}_{\text{deviation around regression line}}$$

Remarkably, the same decomposition is true for the sums of the squared deviations (see KNNL p. 65 for proof).

$$\sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 + \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

Decomposition of sums of squares

$$\sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 + \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

Using the notation from KNNL Equation (2.50)

$$SSTO = SSR + SSE$$

where

- $SSTO$ is called the total sum of squares
- SSR is called the regression sum of squares
- SSE is called the error sum of squares

The larger SSR is in relation to $SSTO$, the larger is the explanatory power of the regression function.

Penguins revisited

```
1 y <- gentoo$body_mass_g
2 ybar <- mean(gentoo$body_mass_g, na.rm = T)
3 lm_fit <- lm(body_mass_g ~ bill_length_mm, data = gentoo)
4 yhat <- lm_fit$fitted.values
5 SST0 <- sum((y - ybar) ^ 2)
6 SSR <- sum((yhat - ybar) ^ 2)
7 SSE <- sum((y - yhat) ^ 2)
8 print(paste0("SST0: ", SST0, ", SSR: ", SSR, ", SSE: ", SSE))
```

```
[1] "SST0: 5884098.36065574, SSR: 900993.675131792, SSE: 4983104.68552394"
```

```
1 print(SSE + SSR)
```

```
[1] 5884098
```

How do we decide if SSR is *large* with respect to $SSTO$, indicating that the regression model explains a large proportion of the total variation?

Degrees of Freedom

Each sum of squares is associated with a number of degrees of freedom.

- $SSTO$ has $n - 1$ degrees of freedom (since we estimate the population mean via the sample mean)
- SSE has $n - 2$ degrees of freedom (since we estimate β_0 and β_1)
- SSR has one degree of freedom. (all $\bar{Y}_i$ are calculated based on one line which has two parameters with one constraint, which is that the deviations sum to zero).

The degrees of freedom are additive:

$$n - 1 = 1 + (n - 2)$$

Mean square terms

Dividing a sum of squares by its degrees of freedom allows us to calculate a mean square term:

The mean square due to regression is:

$$MSR = \frac{SSR}{1} = SSR$$

The mean square due to errors is:

$$MSE = \frac{SSE}{n - 2}$$

Later, we will see how we can use a statistical test to compare MSR and MSE .

Analysis of variance table

After fitting a simple linear regression, we can summarize this decomposition using an **analysis of variance** table, which will typically look something like this:

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Regression	1	SSR	MSR	$F = \frac{MSR}{MSE}$	p -value
Residual	$n - 2$	SSE	MSE		
Total	$n - 1$	$SSTO$			

If we wish to assess the fit of the model, we can perform a hypothesis test using the F -value $F = \frac{MSR}{MSE}$ as a test statistic.

Analysis of variance table: penguins

```
1 lm_fit <- lm(body_mass_g ~ bill_length_mm, data = gentoo)
2 anova(lm_fit)
```

Analysis of Variance Table

Response: body_mass_g

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
bill_length_mm	1	900994	900994	10.668	0.001819 **
Residuals	59	4983105	84459		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Expected mean squares

If MSR is large compared with MSE , then we can say the regression line explains part of the variation in Y . But what counts as *large*?

We have already seen that

$$E(MSE) = \sigma^2$$

In addition (see KNNL p. 69):

$$E(MSR) = \sigma^2 + \beta_1^2 \sum_i (X_i - \bar{X})^2$$

Observe that when $\beta_1 = 0$, $E(MSR) = \sigma^2 = E(MSE)$.

Therefore, if MSR is substantially greater than MSE , this would suggest that $\beta_1 \neq 0$.

F test of $H_0 : \beta_1 = 0$ versus $H_a : \beta_1 \neq 0$

Test statistic. Consider the following test statistic:

$$F^* = \frac{MSR}{MSE}$$

Intuitively, large values of F^* provide support for H_a , while values of F^* near 1 support H_0 .

Sampling distribution of F^* under H_0 . As with any hypothesis test, we compare the observed test statistic with its distribution under the null.

Sampling distribution of F^* under H_0

As a result of Cochran's theorem (KNNL p. 70):

If $Y_1, \dots, Y_n$ are iid $N(\beta_0, \sigma^2)$, so $\beta_1 = 0$, then SSR/σ^2 is $\chi^2(1)$ and SSE/σ^2 is $\chi^2(n-2)$ and the two are independent.

Then

$$F^* = MSR/MSE = \frac{\frac{SSR}{\sigma^2}}{1} \div \frac{\frac{SSE}{\sigma^2}}{n-2}$$

is a ratio of χ^2 variables (each divided by its degrees of freedom).

Sampling distribution of F^* under H_0

In particular, for a simple linear regression model,

$$F^* = MSR/MSE \sim \frac{\chi^2(1)}{1} \div \frac{\chi^2(n-2)}{n-2} \text{ under } H_0$$

A random variable defined in this way follows Snedecor's **F distribution** or the Fisher–Snedecor distribution.

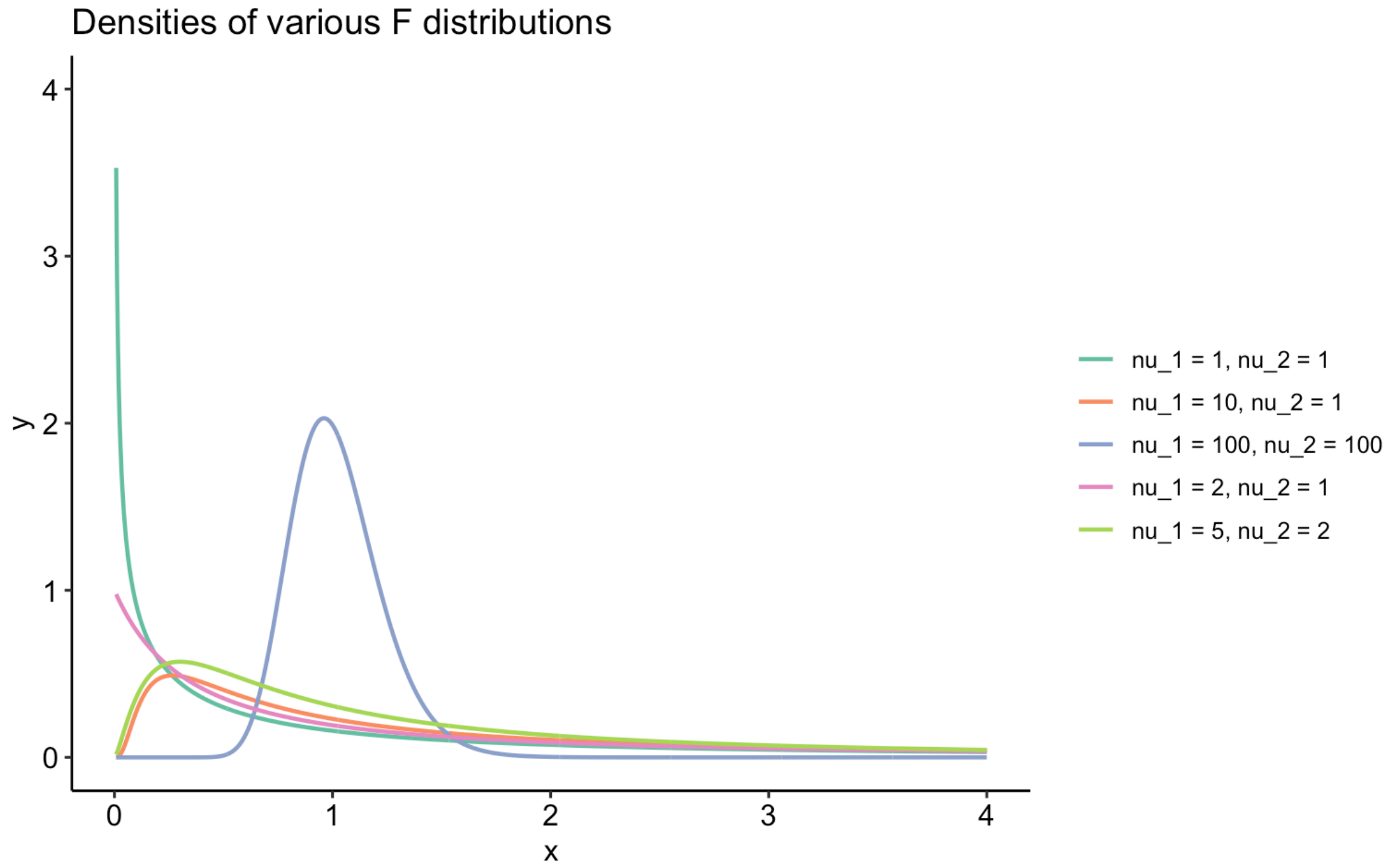
The F-distribution with ν_1 and ν_2 degrees of freedom is the distribution of

$$F(\nu_1, \nu_2) = \frac{U_1/\nu_1}{U_2/\nu_2}$$

where U_1 and U_2 are independent χ^2 variables with ν_1 and ν_2 degrees of freedom, respectively.

F -distribution: Illustration

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***F*-distribution**

The F distribution with ν_1 and ν_2 degrees of freedom has the probability density function

$$f(x) = \frac{\sqrt{\frac{(\nu_1 x)^{\nu_1} \nu_2^{\nu_2}}{(\nu_1 x + \nu_2)^{\nu_1 + \nu_2}}}}{x \text{B}\left(\frac{\nu_1}{2}, \frac{\nu_2}{2}\right)}$$

where B denotes the beta function:

$$\text{B}(z_1, z_2) = \int_0^1 t^{z_1-1} (1-t)^{z_2-1} dt$$

If $X \sim F(\nu_1, \nu_2)$, then

$$E(X) = \frac{\nu_2}{\nu_2 - 2} \text{ when } \nu_2 > 2$$

$$\text{Var}(X) = \frac{2 \nu_2^2 (\nu_1 + \nu_2 - 2)}{\nu_1 (\nu_2 - 2)^2 (\nu_2 - 4)} \text{ when } \nu_2 > 4$$

Decision rule for F -test

We use an upper-tail test that compares F^* with the null sampling distribution $F(1, n - 2)$. Let $F_{1-\alpha;1,n-2}$ be the $(1 - \alpha)100\%$ percentile of the $F(1, n - 2)$ distribution.

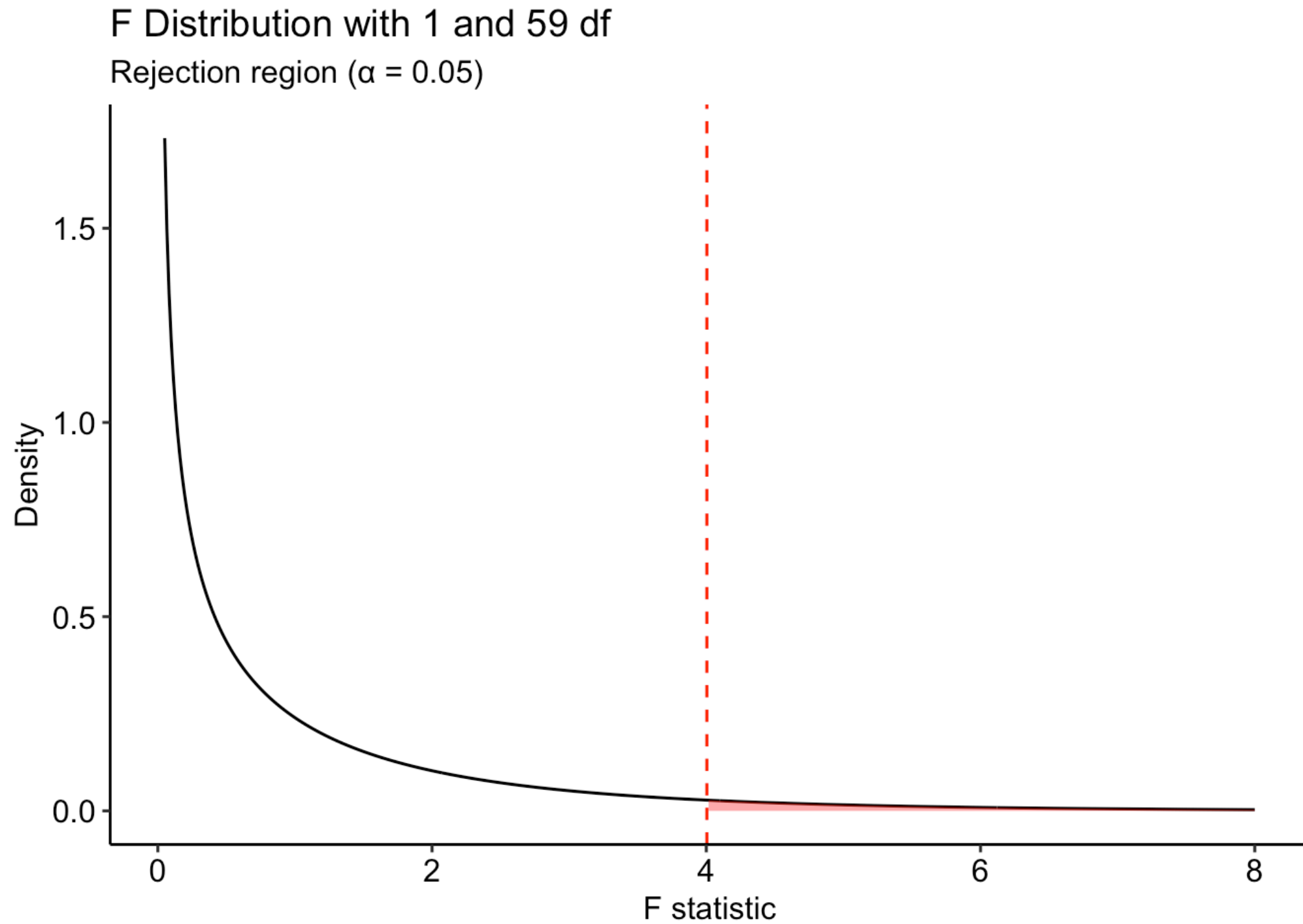
Then, the decision rule is

If $F^* \leq F_{1-\alpha;1,n-2}$, fail to reject H_0 ,

If $F^* > F_{1-\alpha;1,n-2}$, reject H_0

Penguins example, illustrated

► Code



Penguins example, illustrated

```
1 lm_fit <- lm(body_mass_g ~ bill_length_mm, data = gentoo)
2 anova(lm_fit)
```

Analysis of Variance Table

Response: body_mass_g

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
bill_length_mm	1	900994	900994	10.668	0.001819 **
Residuals	59	4983105	84459		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Penguins example, illustrated

Note that for simple linear regression, this is equivalent to the t -test for $\beta_1 = 0$ vs. $\beta_1 \neq 0$ we saw earlier (in fact, $F^* = (t^*)^2$):

```
1 summary(lm_fit)
```

Call:

```
lm(formula = body_mass_g ~ bill_length_mm, data = gentoo)
```

Residuals:

Min	1Q	Median	3Q	Max
-740.5	-171.1	-6.3	182.4	827.5

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	3256.42	683.29	4.766	1.27e-05	***
bill_length_mm	45.04	13.79	3.266	0.00182	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 290.6 on 59 degrees of freedom

Multiple R-squared: 0.1531, Adjusted R-squared: 0.1388

F-statistic: 10.67 on 1 and 59 DF. p-value: 0.001819

Coefficient of determination

There is, unfortunately, no single measure that describes whether a regression model is useful. Usefulness is always context dependent: ex. how accurate do your predictions need to be?

Nevertheless, you will often see the R^2 value reported as a measure of the model's explanatory power.

R^2 is called the **coefficient of determination** and is defined as the proportion of variation explained by the regression model:

$$R^2 = \frac{SSR}{SSTO} = 1 - \frac{SSE}{SSTO}$$

Coefficient of determination: properties

1. Since $0 \leq SSE \leq SSTO$, $0 \leq R^2 \leq 1$.
2. When all observations fall on the regression line, $SSE = 0$ and $R^2 = 1$.
3. When the fitted line is horizontal ($\beta_1 = 0$), $SSTO = SSR$ and $R^2 = 0$.

Coefficient of determination: pitfalls

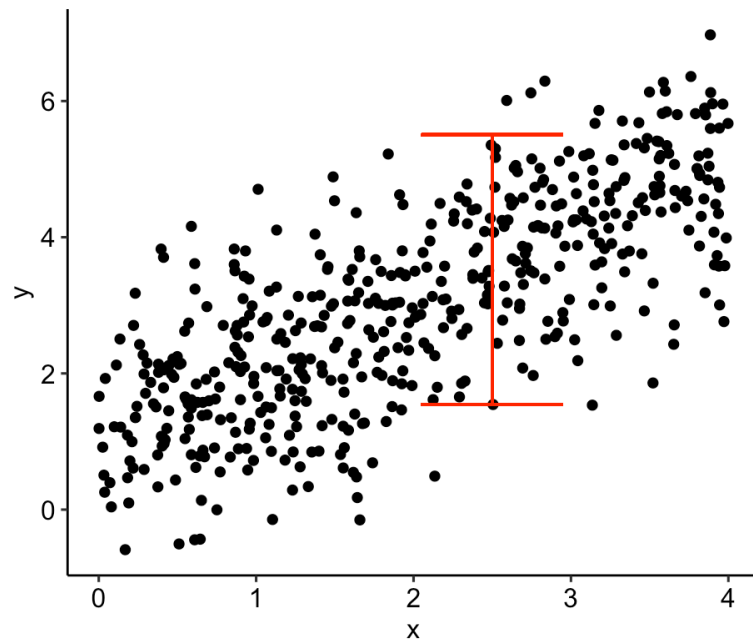
1. High R^2 does not necessarily mean we can make useful predictions.
2. High R^2 does not necessarily mean the estimated regression line is a good fit.
3. Low R^2 does not necessarily mean that X and Y are not related.

Coefficient of determination: pitfalls

► Code

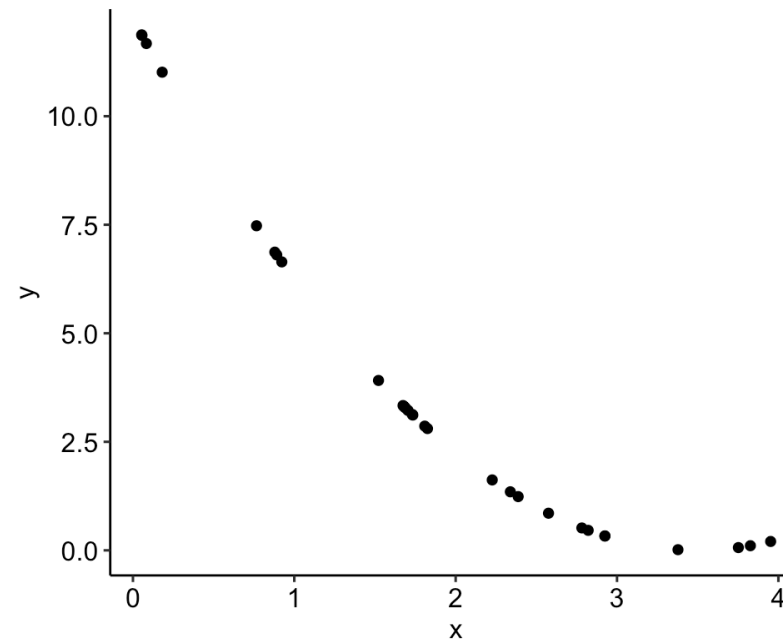
High R-squared can still yield
wide prediction intervals

R-squared = 0.567



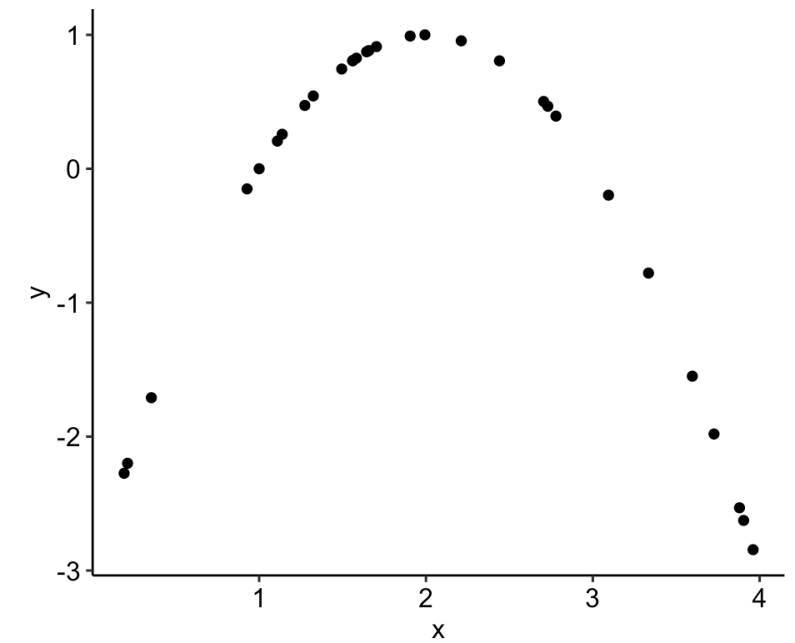
High R-squared can hide
non-linear associations

R-squared = 0.865



Low R-squared can hide
non-linear associations

R-squared = 0.097



Coefficient of correlation

R^2 is of course related to the **correlation coefficient**, which quantifies the strength of the **linear** association between X and Y :

$$r = \pm \sqrt{R^2}$$

where the sign depends on the direction of the association (negative sign for negative association, and vice versa).

Class outline

- Assumptions, diagnostics, and model evaluation
- Discuss properties of residuals
- Possible remedial measure
- Coding exercises

Review: simple linear regression model

Recall the **simple** linear regression with one outcome and one predictor. The model is typically written out:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

where:

- Y_i is the value of the i th outcome
- β_0 is the intercept parameter and β_1 is the slope parameter
- X_i is a **known** constant representing the i th predictor
- ε_i is a random error with mean zero and finite variance σ^2 ; all errors are independent, so if $i \neq j$, ε_i and ε_j are independent (in certain cases we may want to assume normality of ε_i)
- $i = 1, \dots, n$

Assumptions of simple linear regression

In decreasing order of importance (ROS Ch. 11):

1. **Validity.** The variables used should accurately reflect the quantities of interest, the model should include all relevant predictors, and the model should be applicable for the research question.

Check:

- *Is the data appropriate for the research question?* ex. If the goal is to study the effect of sleep on cognitive development, data on grades in school may not give the full picture.
- *Is regression necessary to answer your research question?* ex. If you wish to show that prices have increased over time, a plot of the Consumer Price Index over time may be sufficient.

Assumptions of simple linear regression

2. **Representativeness.** The sample data is representative of the population of interest (after conditioning on the predictor variables).

Check:

- *Is the sample systematically different from the population, in a way that we can't include in our model?* For example, if we wish to study the association between height and earnings, it may be okay to have an overrepresentation of taller people. However, if older people are overrepresented (and the relationship between Y and X varies across age), then our model may be unreasonable.
- *If we are making predictions, is the new case like the existing data?* If we wish to make predictions for a future event (like an election), our model will only be useful if the future case resembles the past cases.

Assumptions of simple linear regression

3. **Additivity and linearity.** The mean function must be correct:

$$E(Y_i | X_i) = \beta_0 + \beta_1 X_i$$

Check:

- *Does a linear model make sense for the mean relationship between Y and X ? Would a nonlinear function make more sense? For example $E(Y_i | X_i) = \exp(\beta_1 X_i)$. In some cases, transformation can be used to create a linear relationship.*

Assumptions of simple linear regression

4. **Independence of errors.** The simple linear regression model assumes errors are independent, which may be violated when data are correlated in time or space, or when observations are clustered.
5. **Equal variance of errors.** Unequal error variance (heteroskedasticity) is a common issue that affects inference with the regression model.

These issues may not necessarily prevent us from unbiased estimation of the regression line, but if these assumptions are broken, OLS may be inefficient, meaning the parameter estimates are not optimal in terms of variance.

Assumptions of simple linear regression

6. **Normality of errors.** Normality of errors enables inference with small samples, enabling us to conduct hypothesis tests even when the sample is small. However, for the purpose of estimating the regression line, normality is not critical. For this reason, we will not worry so much about the normality of errors.

Simple linear regression, step by step

1. Starting with a research question, propose a regression analysis. Identify the dataset and the predictor / outcome variables. Identify the goal(s): Inference on β_1 ? Prediction of new cases? Identify the necessary assumptions and assess whether any issues are likely to arise.

ex. Propose a strategy for using regression to answer the question: *Are people with shorter work commutes happier?*

ex. Propose a strategy for using regression to answer the question: *Can we predict students' college grades based on their SAT scores?*

Simple linear regression, step by step

2. Clean the data and then summarize / visualize the predictor and outcome variables.
3. Fit the linear regression and assess the model fit.
4. If everything seems satisfactory, use the fitted model to answer your research question.
5. Present your analysis, describing all steps necessary to reproduce your results. Discuss any caveats or limitations of your analysis.

Your friend, the scatter plot

What can we say about the fit of the following regression model?

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```
Call:
lm(formula = y ~ x, data = df)

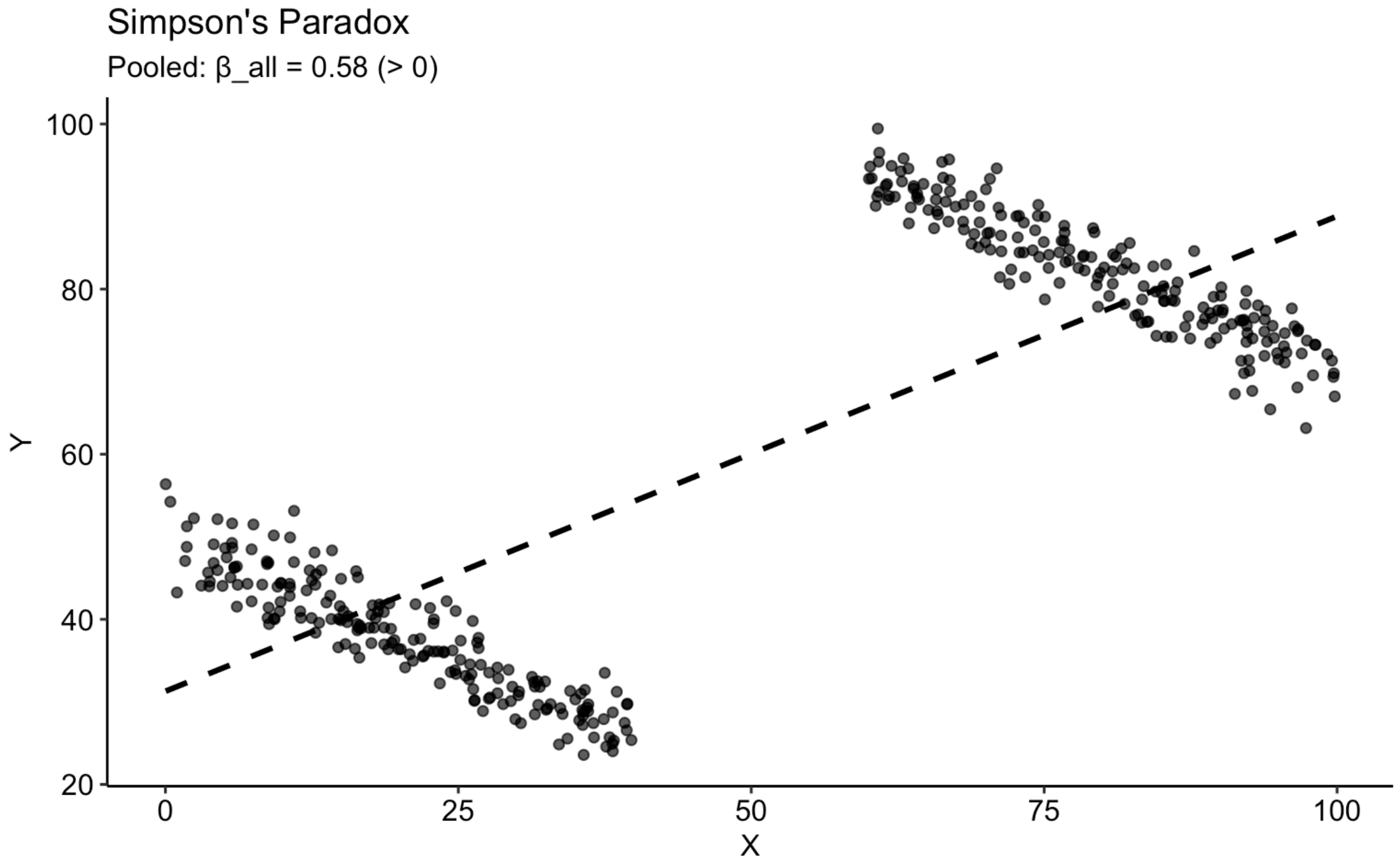
Residuals:
    Min       1Q   Median       3Q      Max
-29.237 -11.618  -0.533   11.013   33.169

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 31.28304    1.36194   22.97  <2e-16 ***
x           0.57551    0.02298   25.04  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 14.62 on 398 degrees of freedom
Multiple R-squared:  0.6118,    Adjusted R-squared:  0.6108
F-statistic: 627.1 on 1 and 398 DF,  p-value: < 2.2e-16
```

Your friend, the scatter plot

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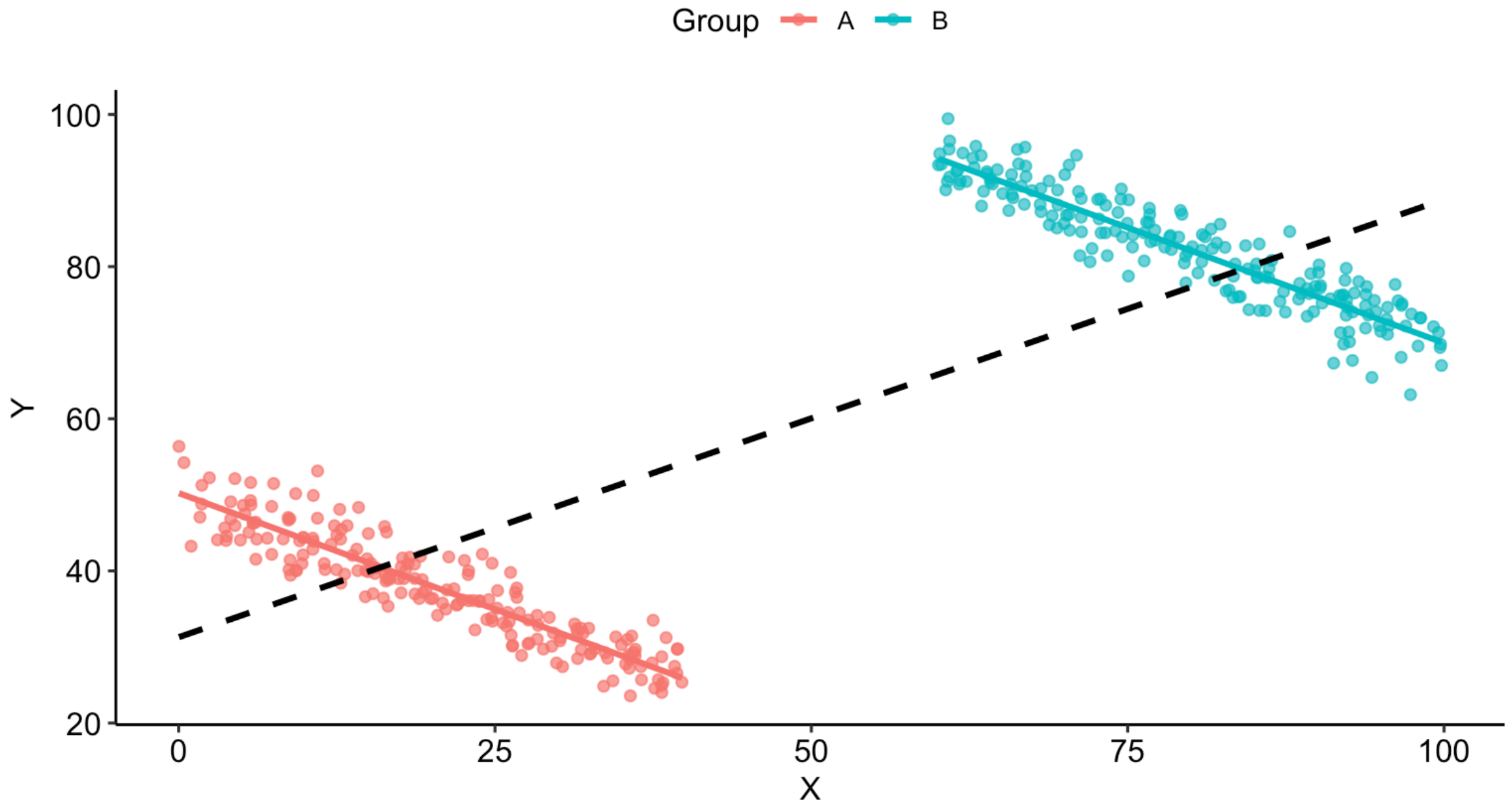


Your friend, the scatter plot

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Simpson's Paradox

Within groups: $\beta_A = -0.61$, $\beta_B = -0.61$ (both < 0); Pooled: $\beta_{\text{all}} = 0.58$ (> 0)



Residuals

Recall that the residual e_i is the difference between the observed and predicted values:

$$e_i = Y_i - \hat{Y}_i$$

Contrast this with the unknown true error $\varepsilon_i = Y_i - E(Y_i)$.

Examining residuals is a common way to assess model fit: if the model is correct, then the residuals should be close (though not equal) to the true errors.

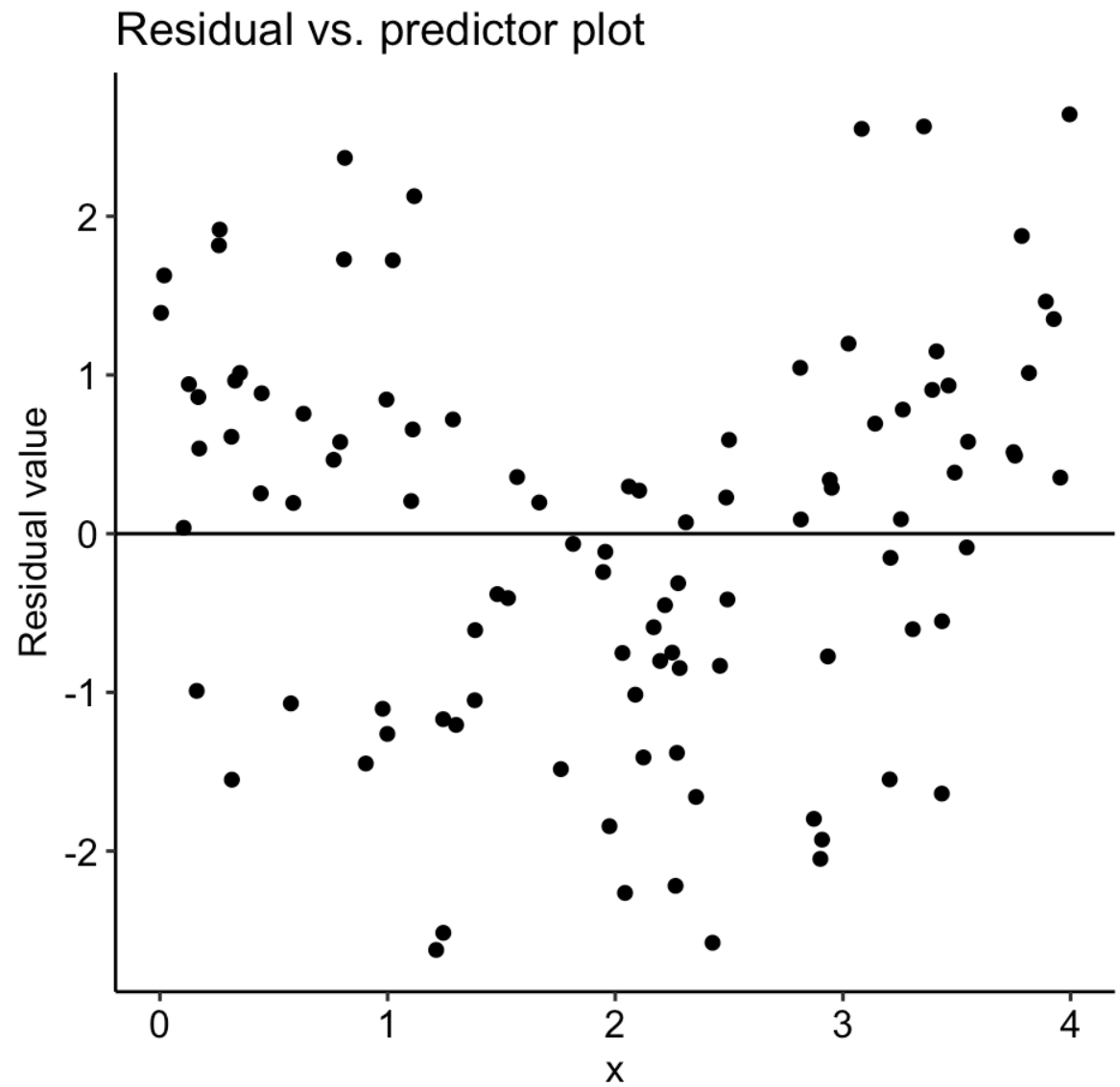
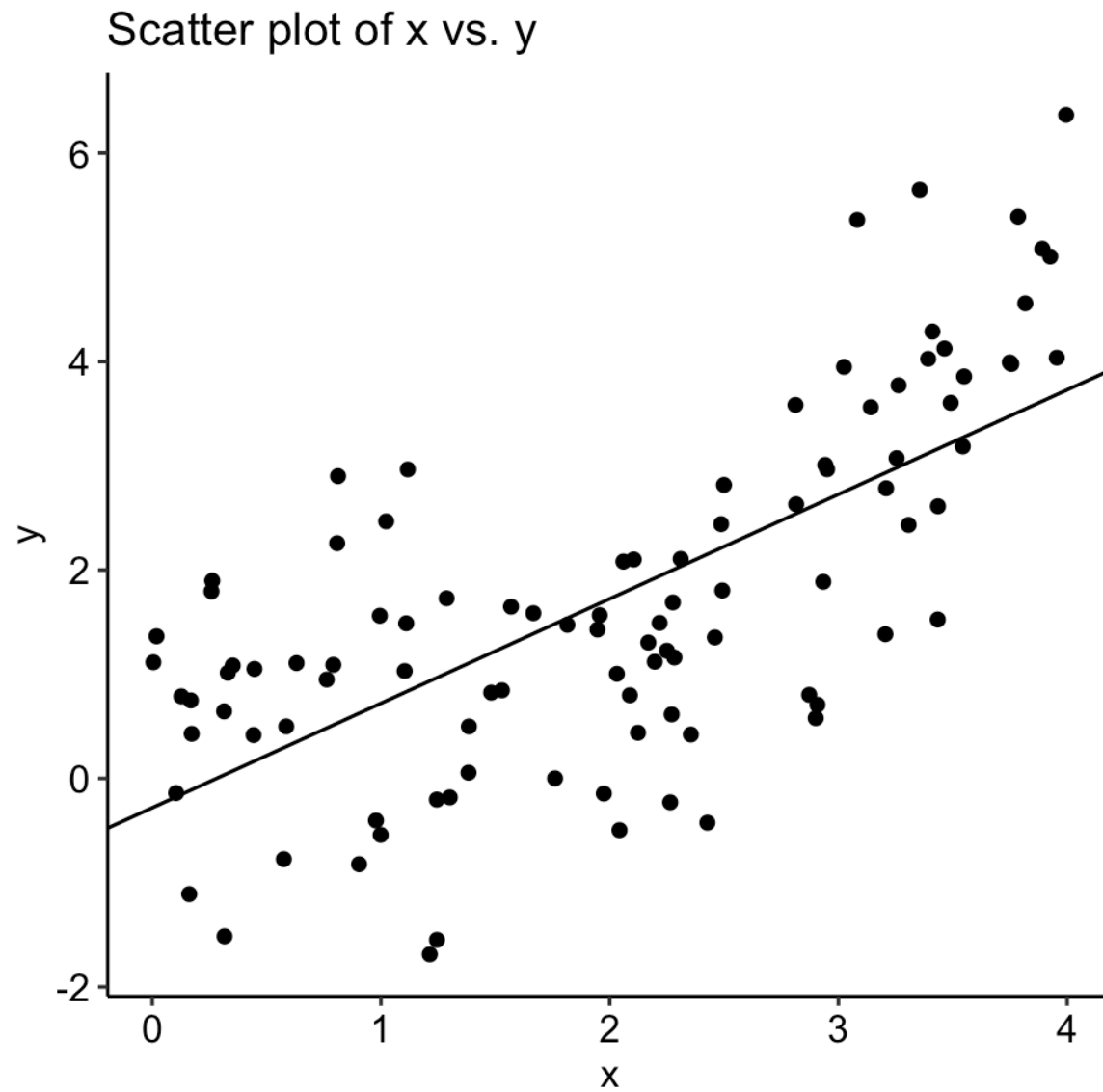
Assessing model fit: residuals

KNNL Chapter 3.2 identifies six common departures from the simple linear regression model that we can assess by examining the residuals:

1. The regression function is not linear.
2. The error terms do not have equal variance.
3. The error terms are not independent.
4. The model fits all but one or a few outliers.
5. The error terms are not normally distributed.
6. One or several important predictors have been omitted from the model.

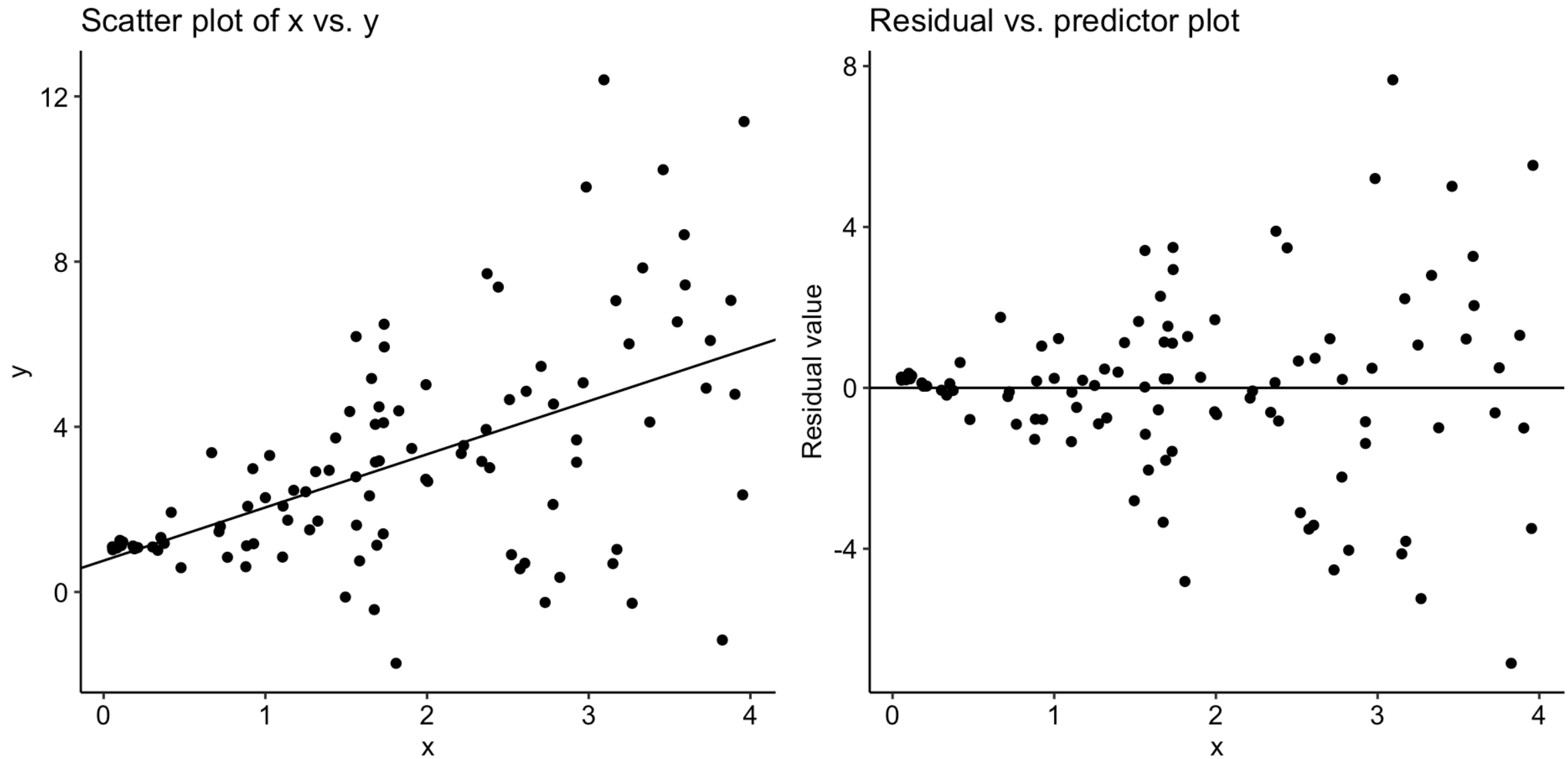
The regression function is not linear

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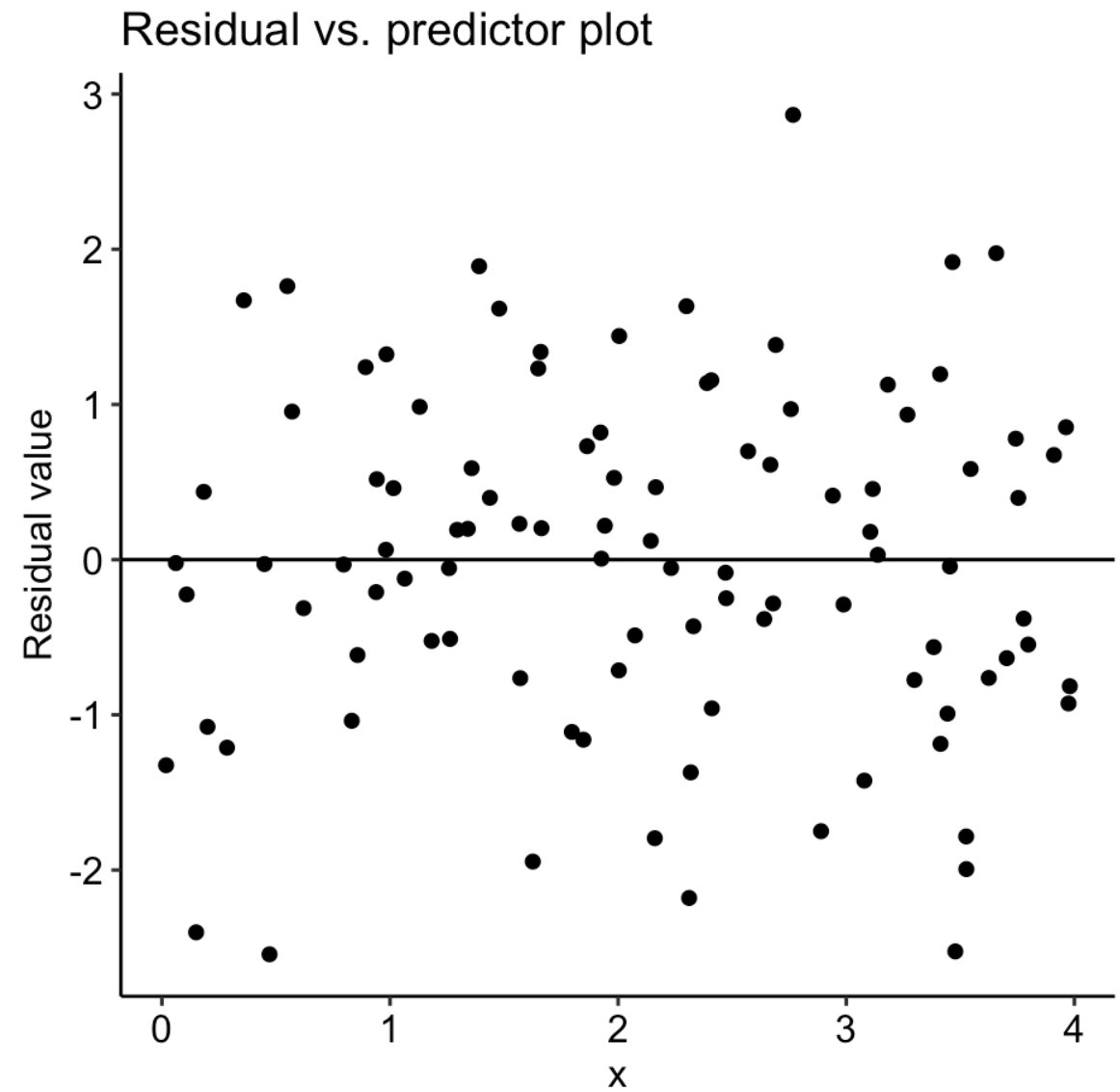
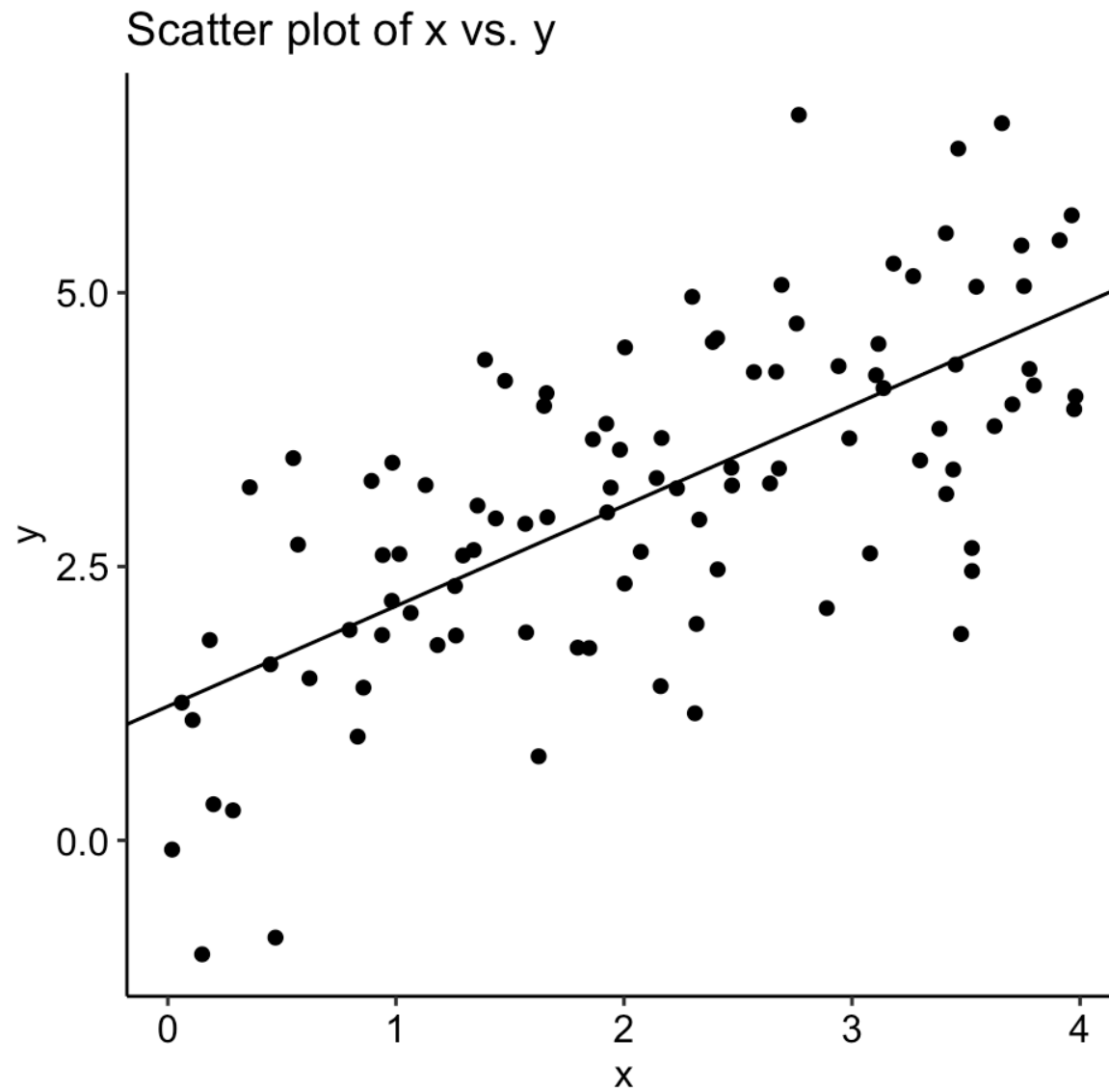
The error terms do not have equal variance

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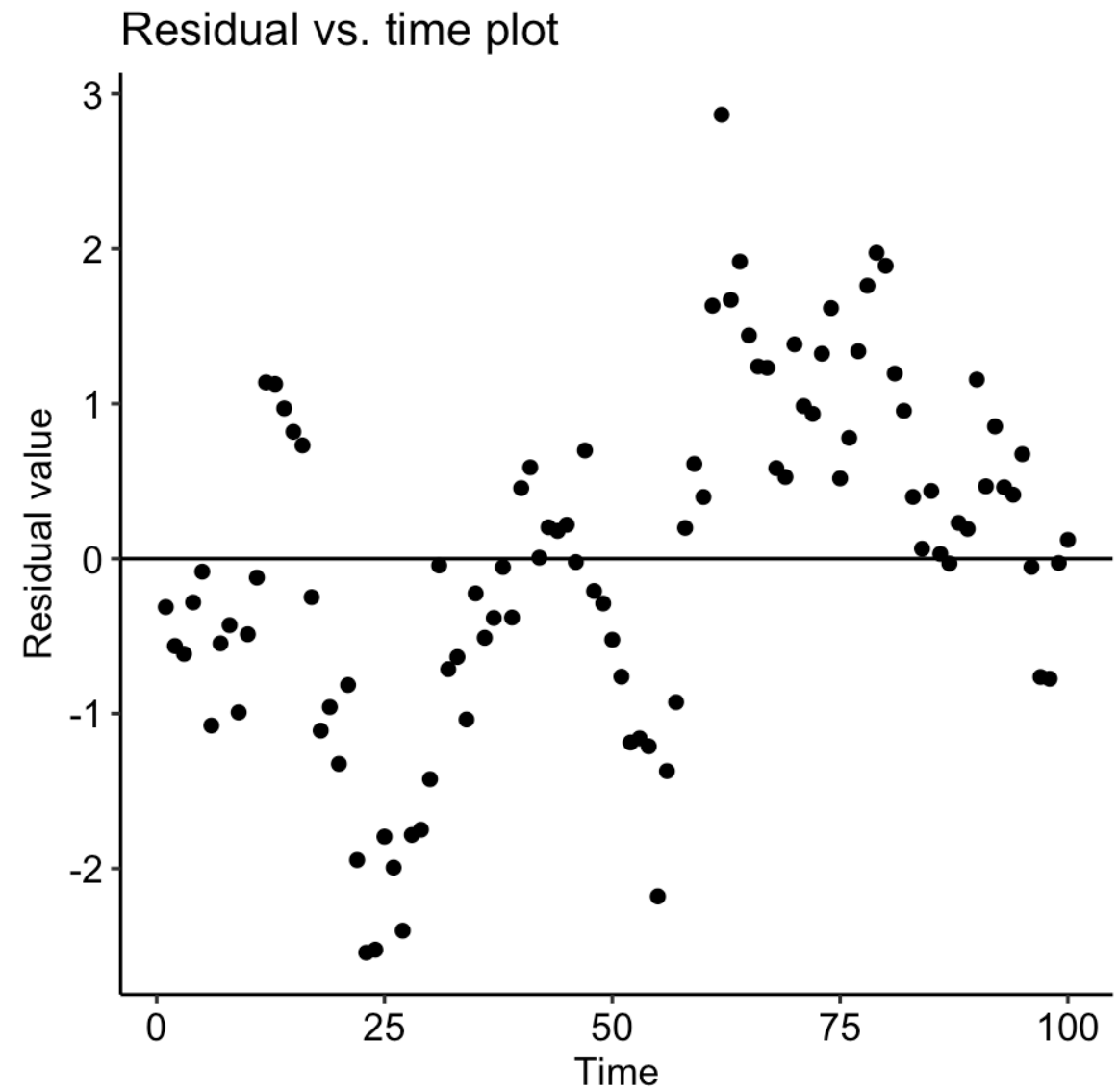
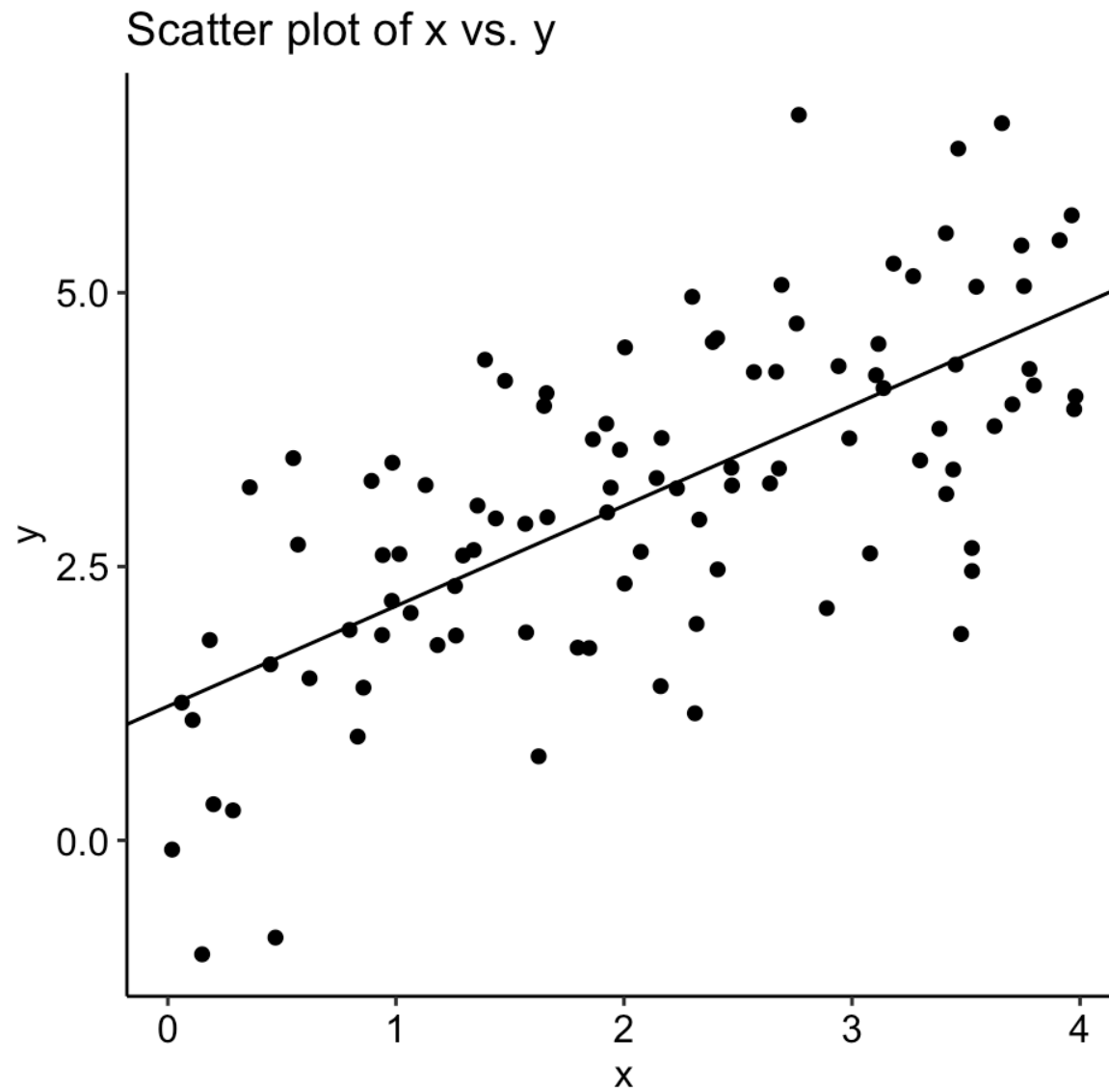
The error terms are not independent

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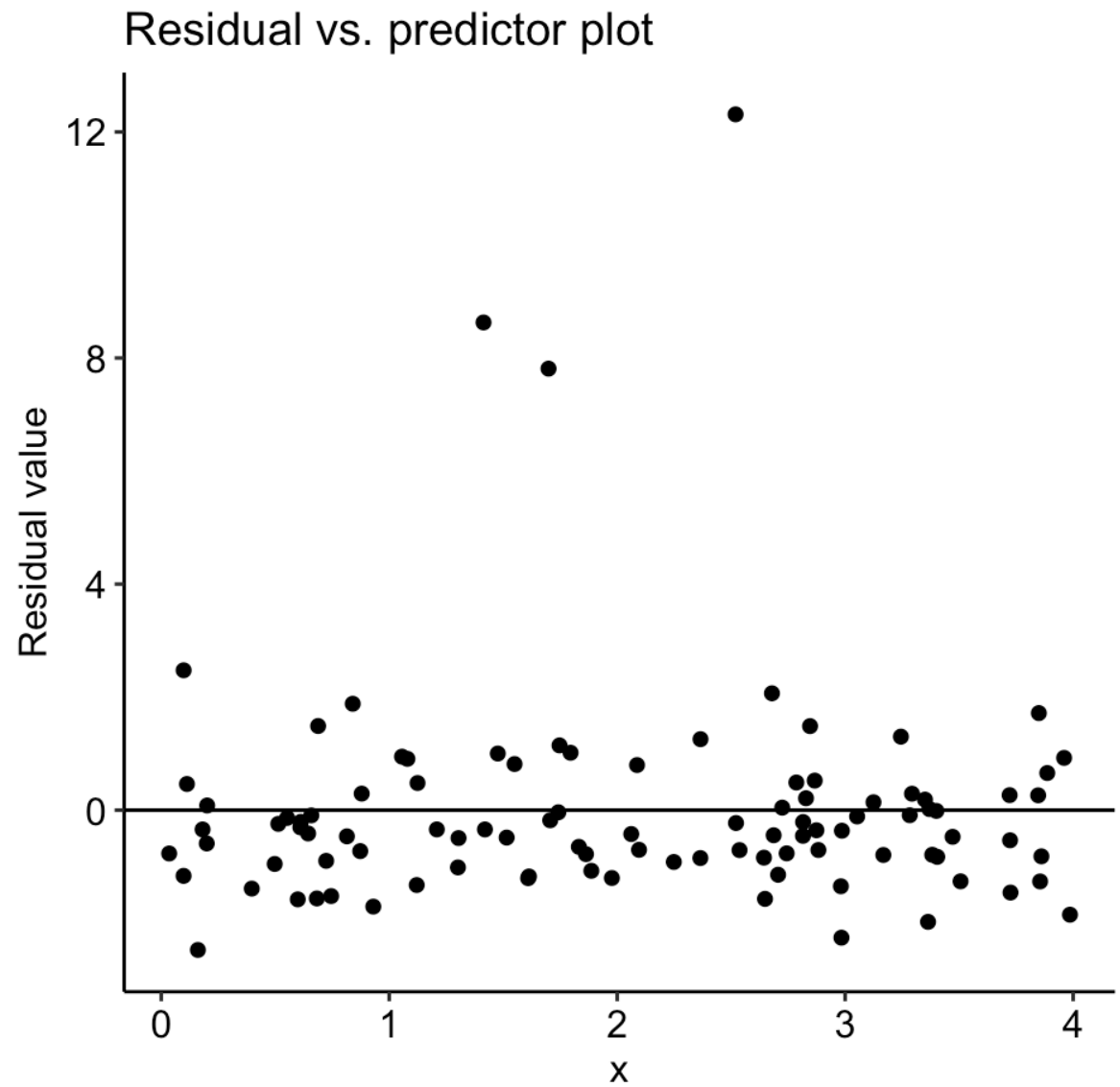
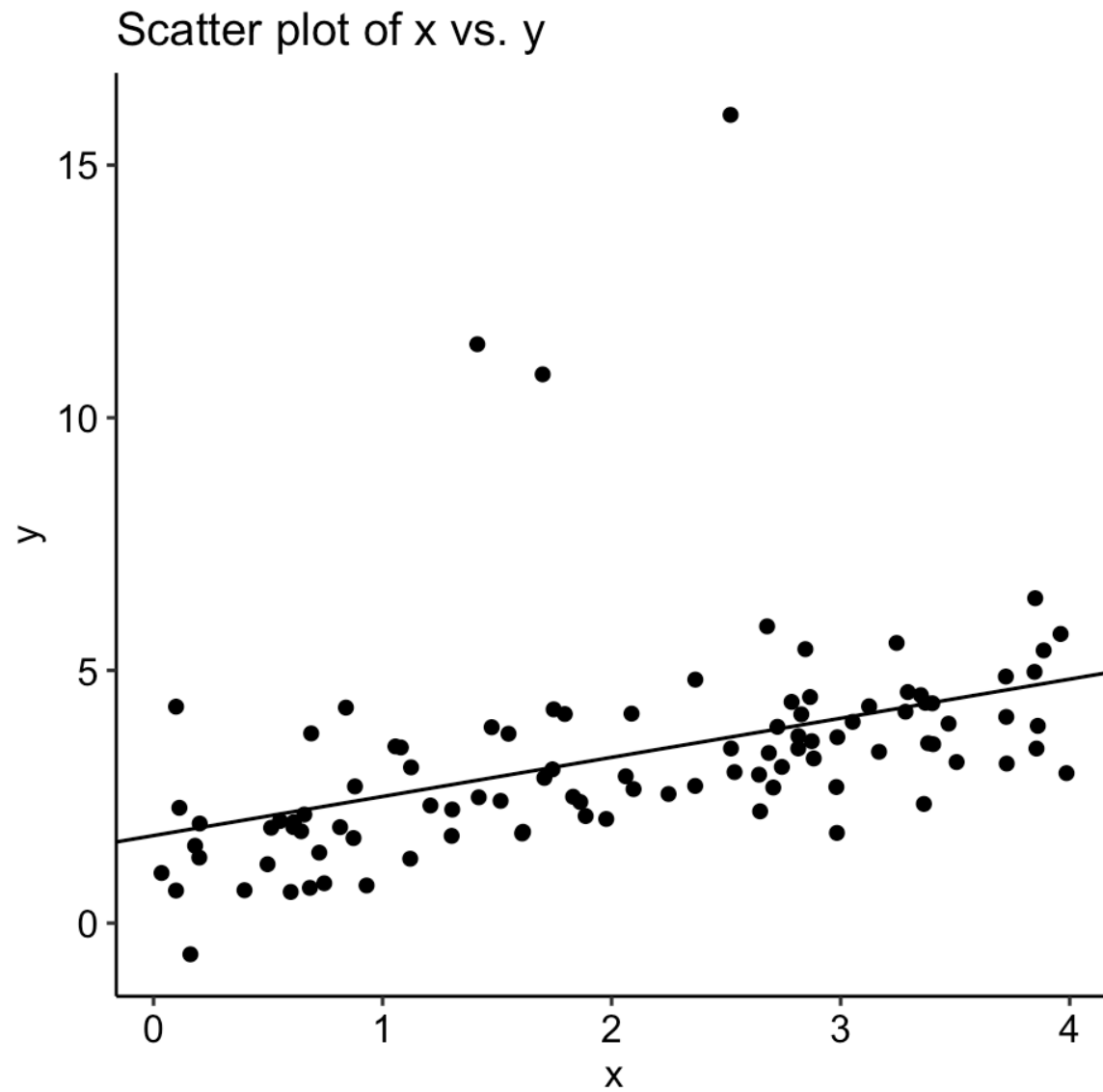
The error terms are not independent

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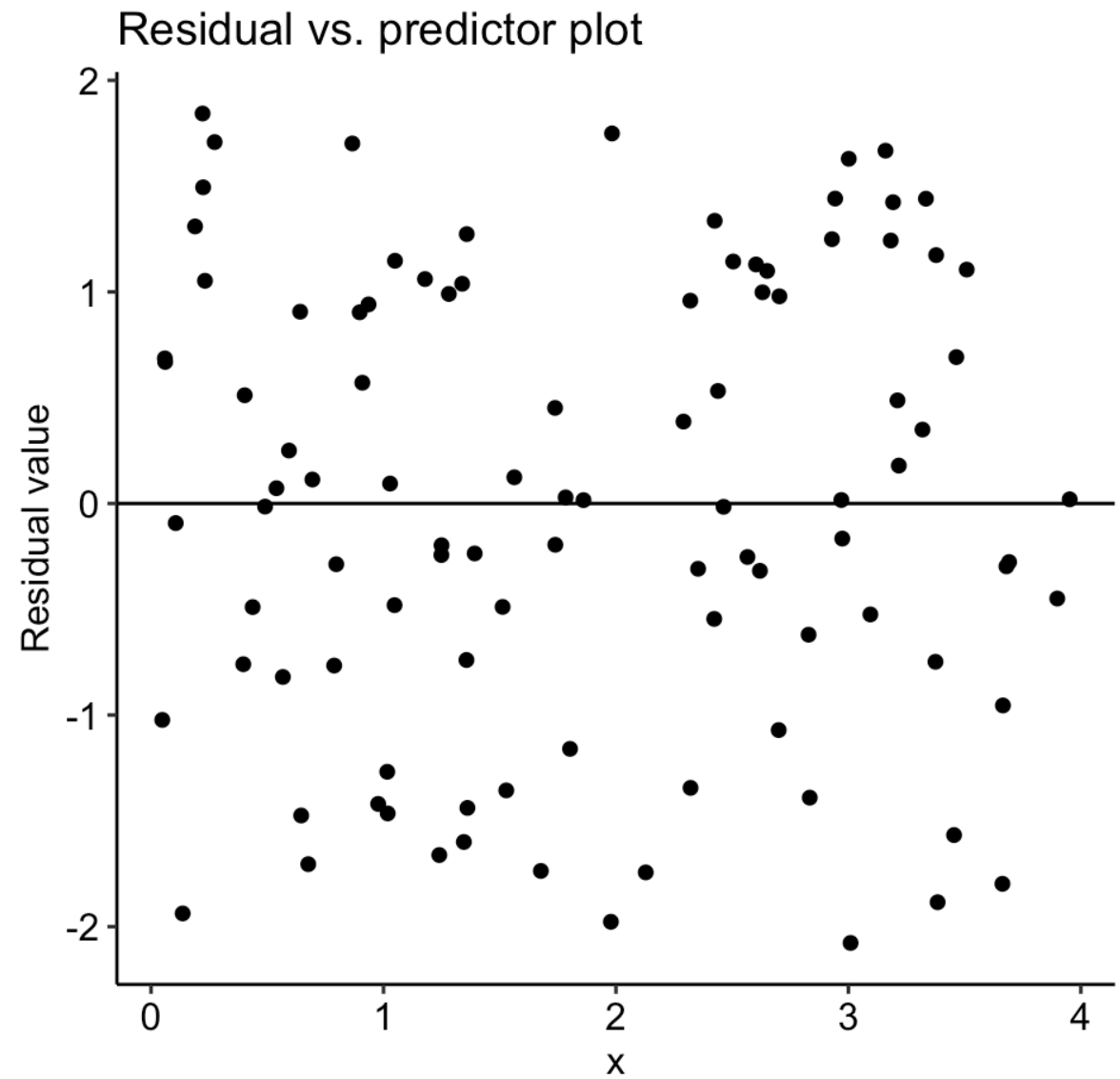
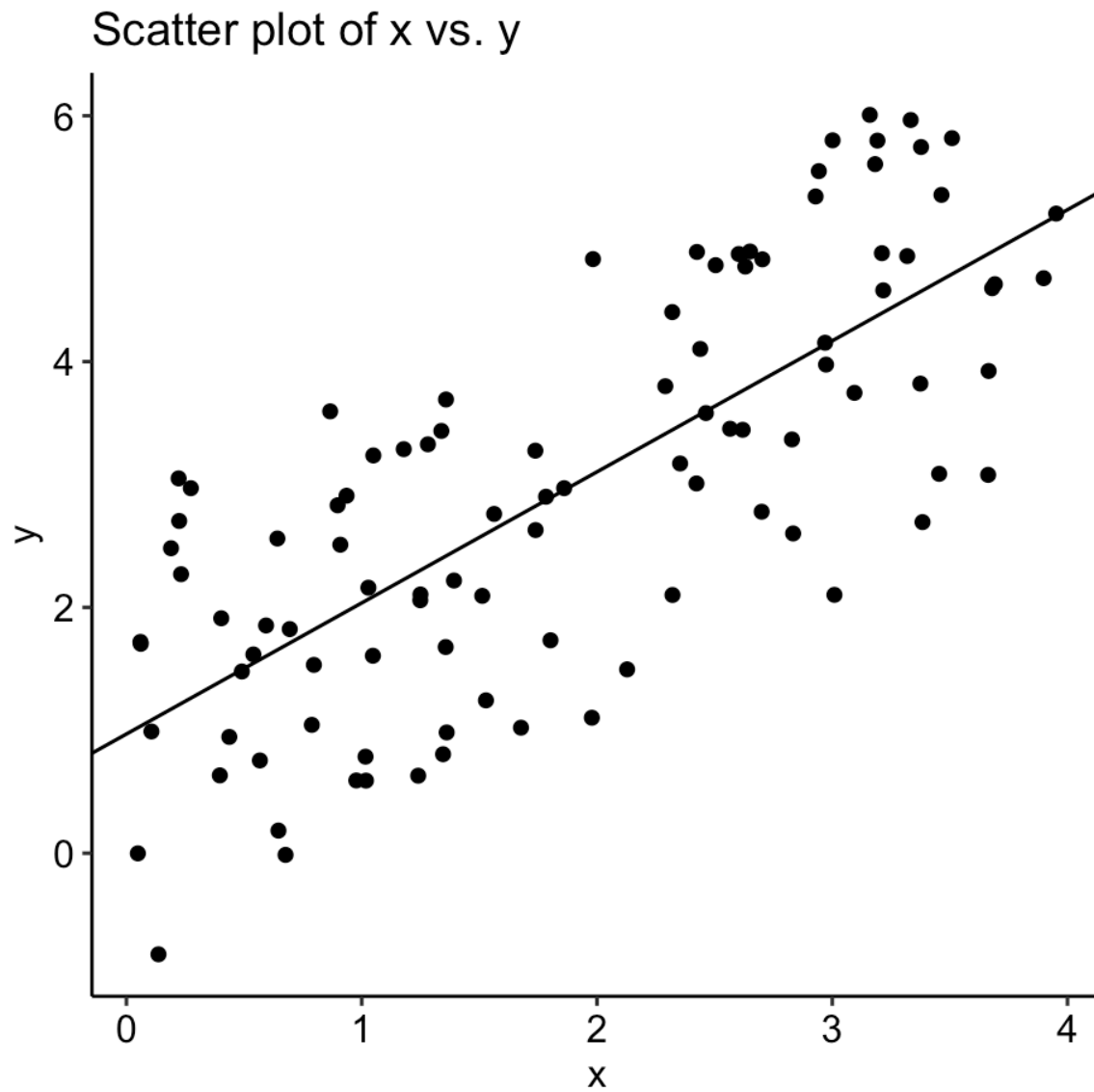
The model fits all but one or a few outliers

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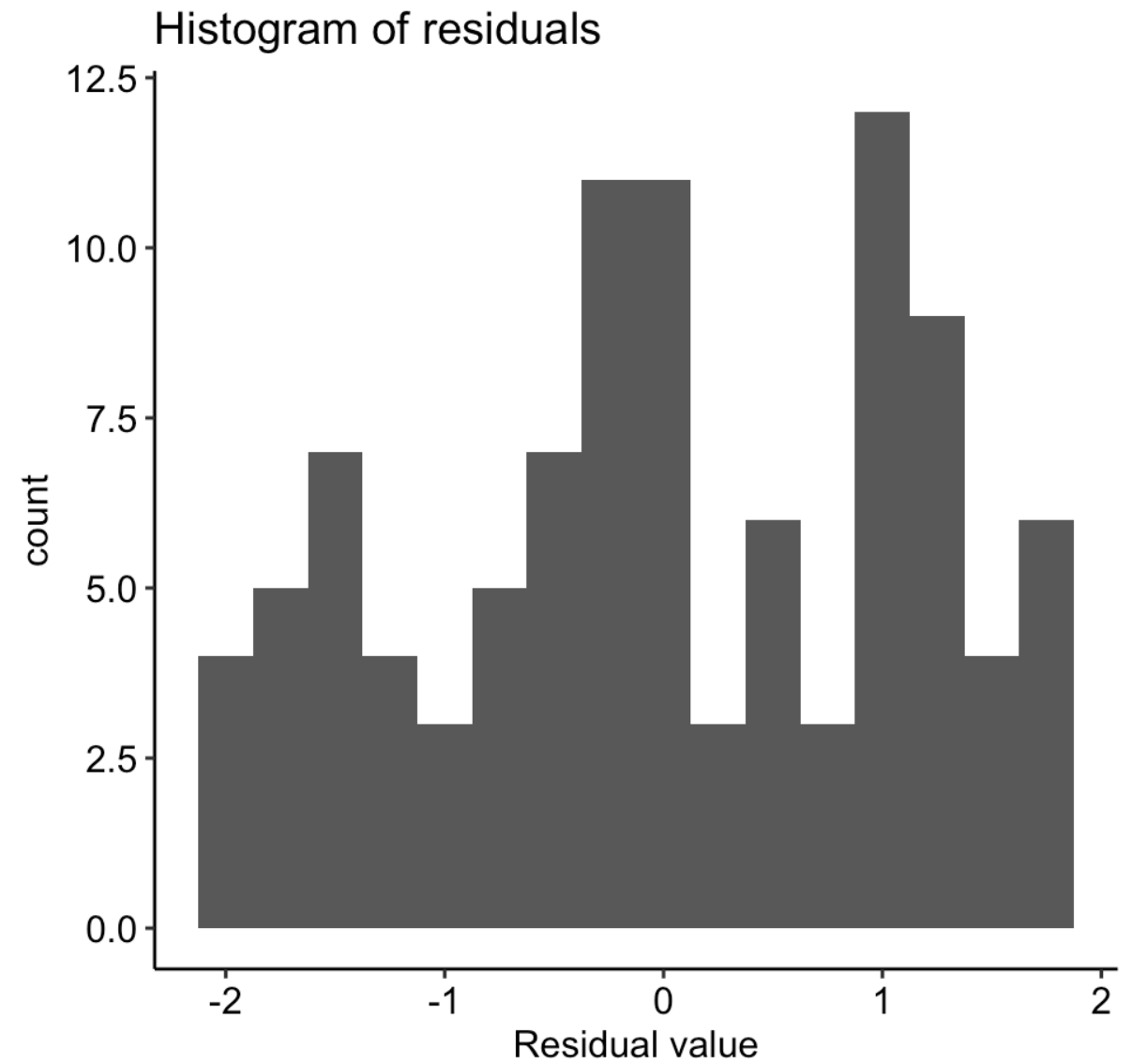
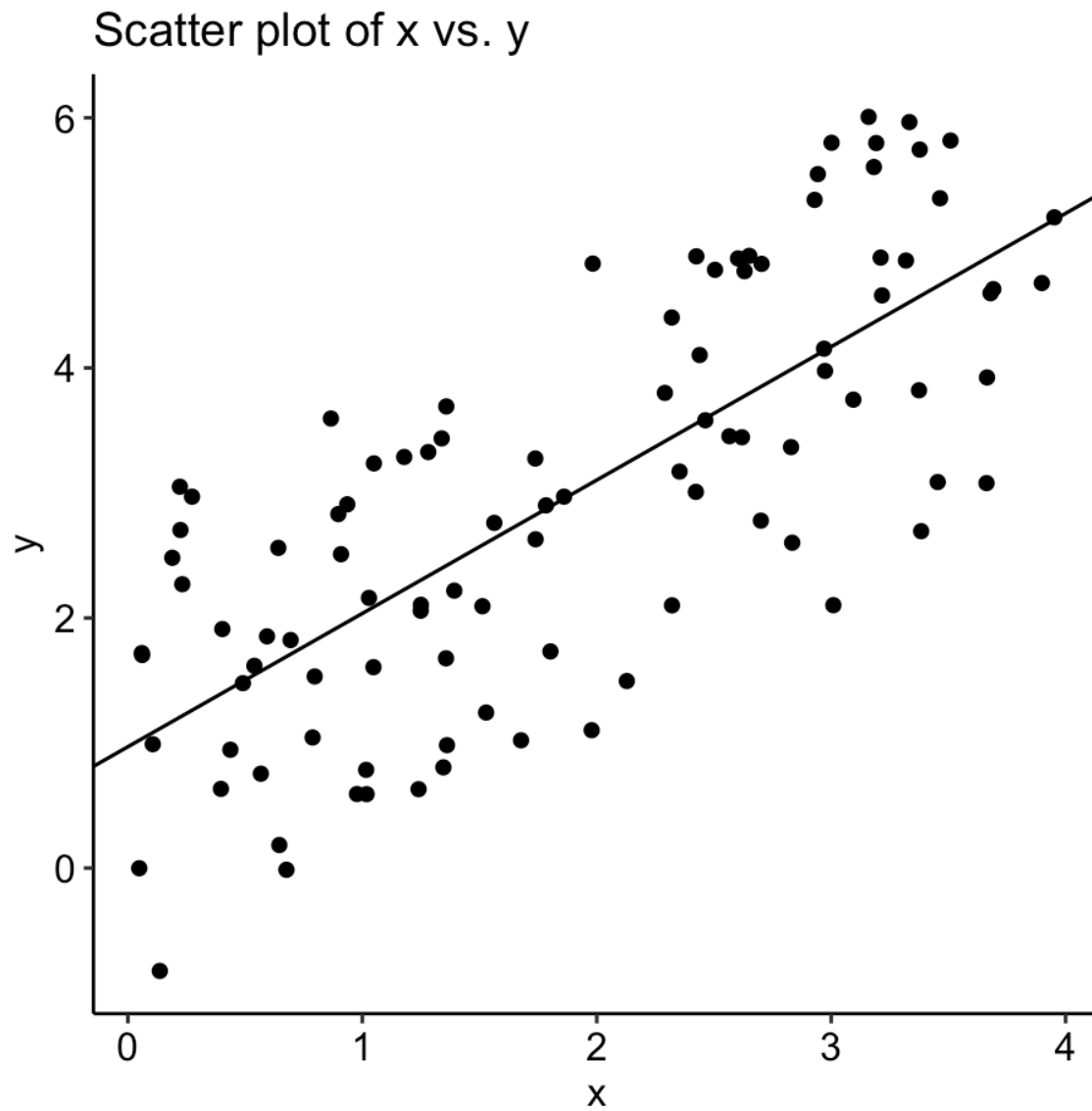
The error terms are not normally distributed

► Code



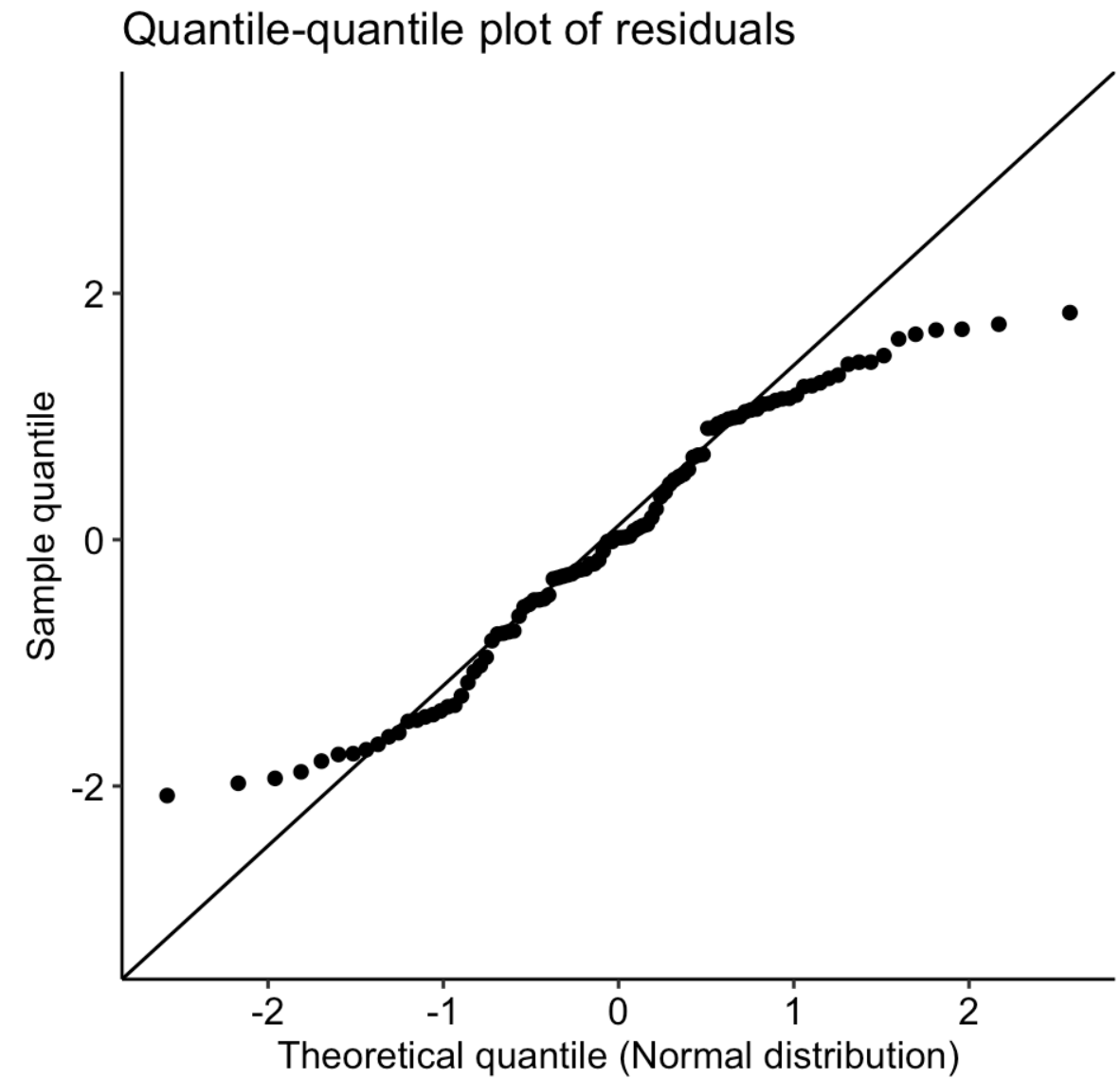
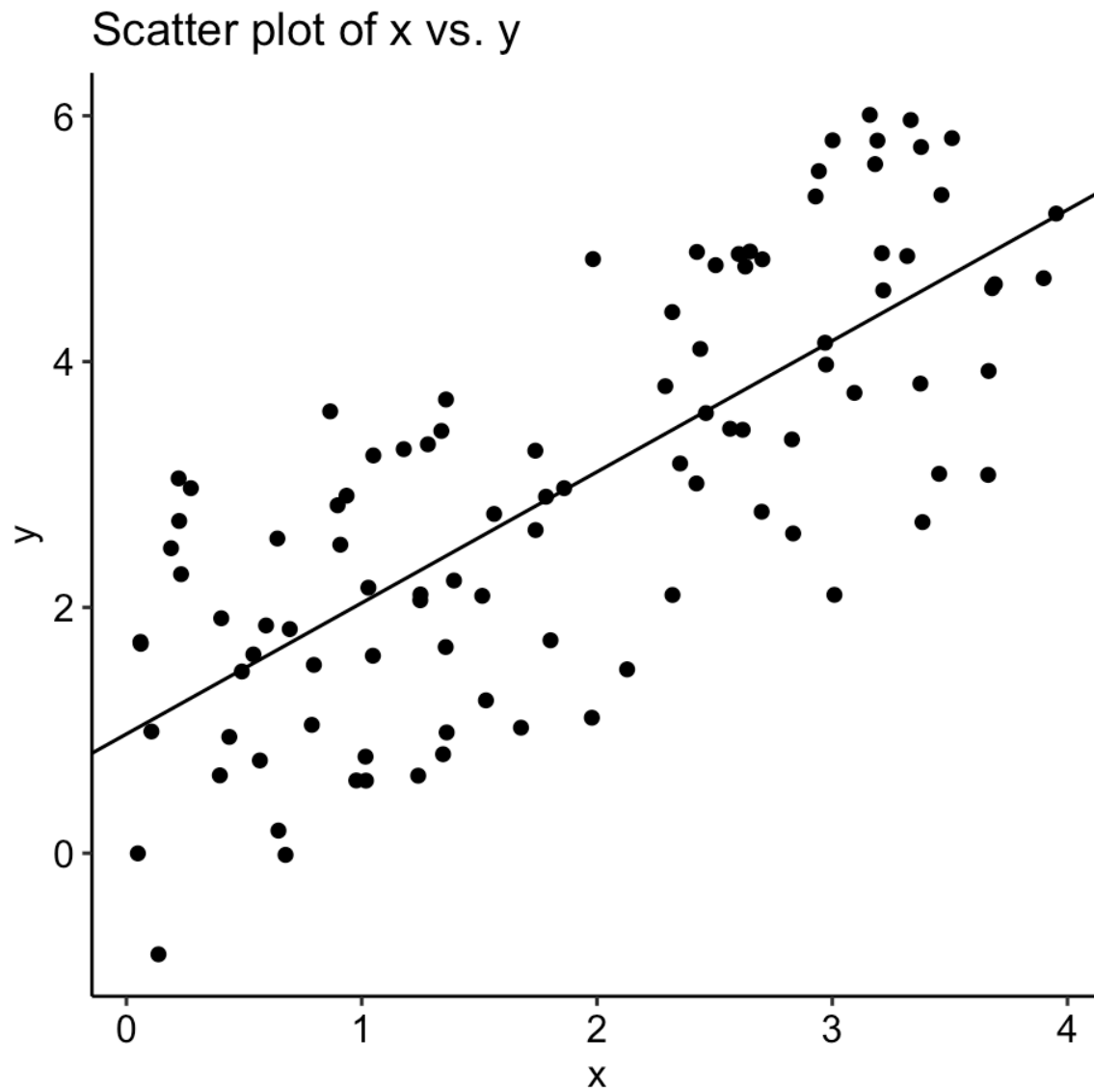
The error terms are not normally distributed

► Code



The error terms are not normally distributed

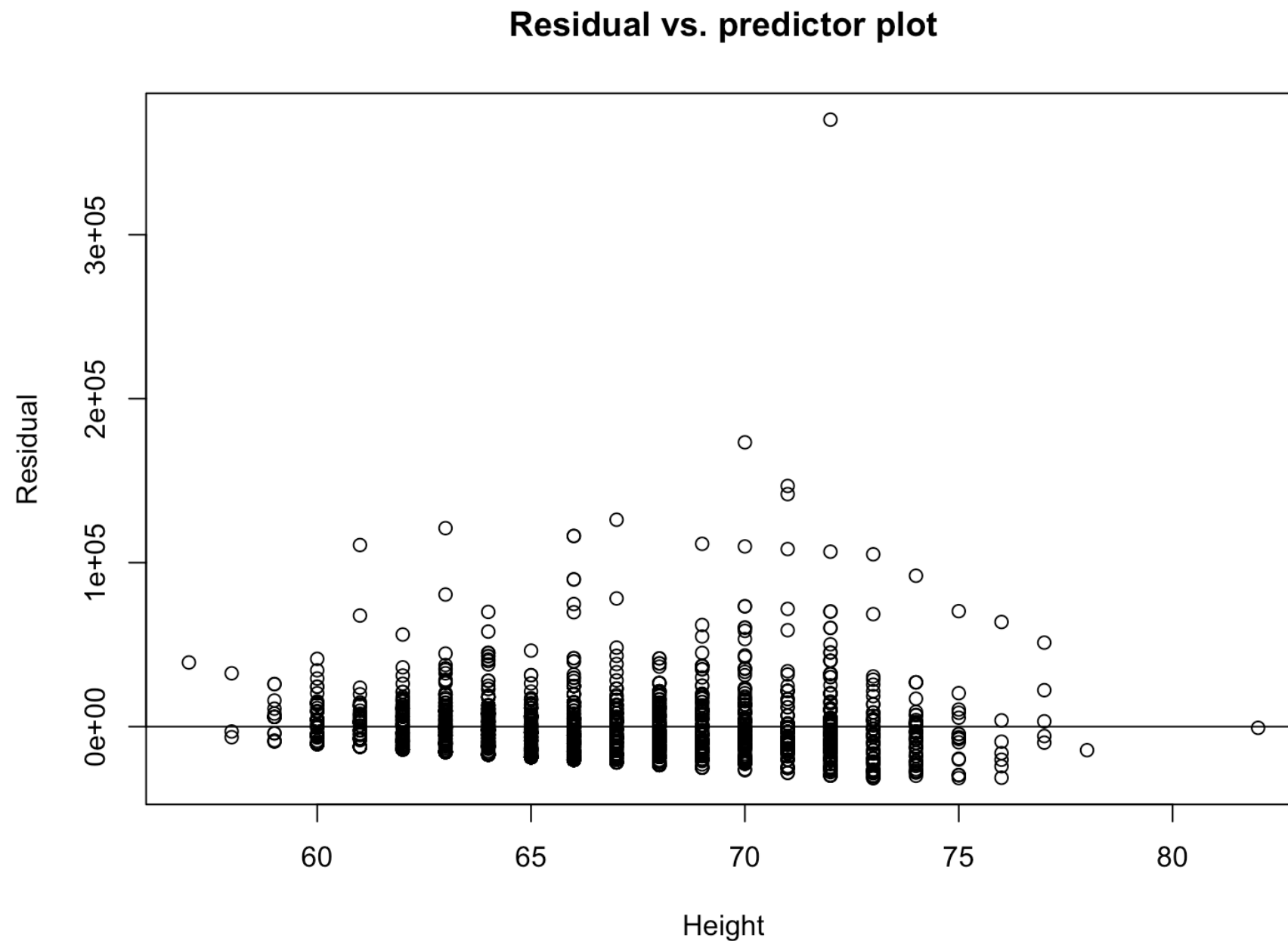
► Code



Assessing model fit: example

Do you notice any issues with the following residual vs. predictor plot?

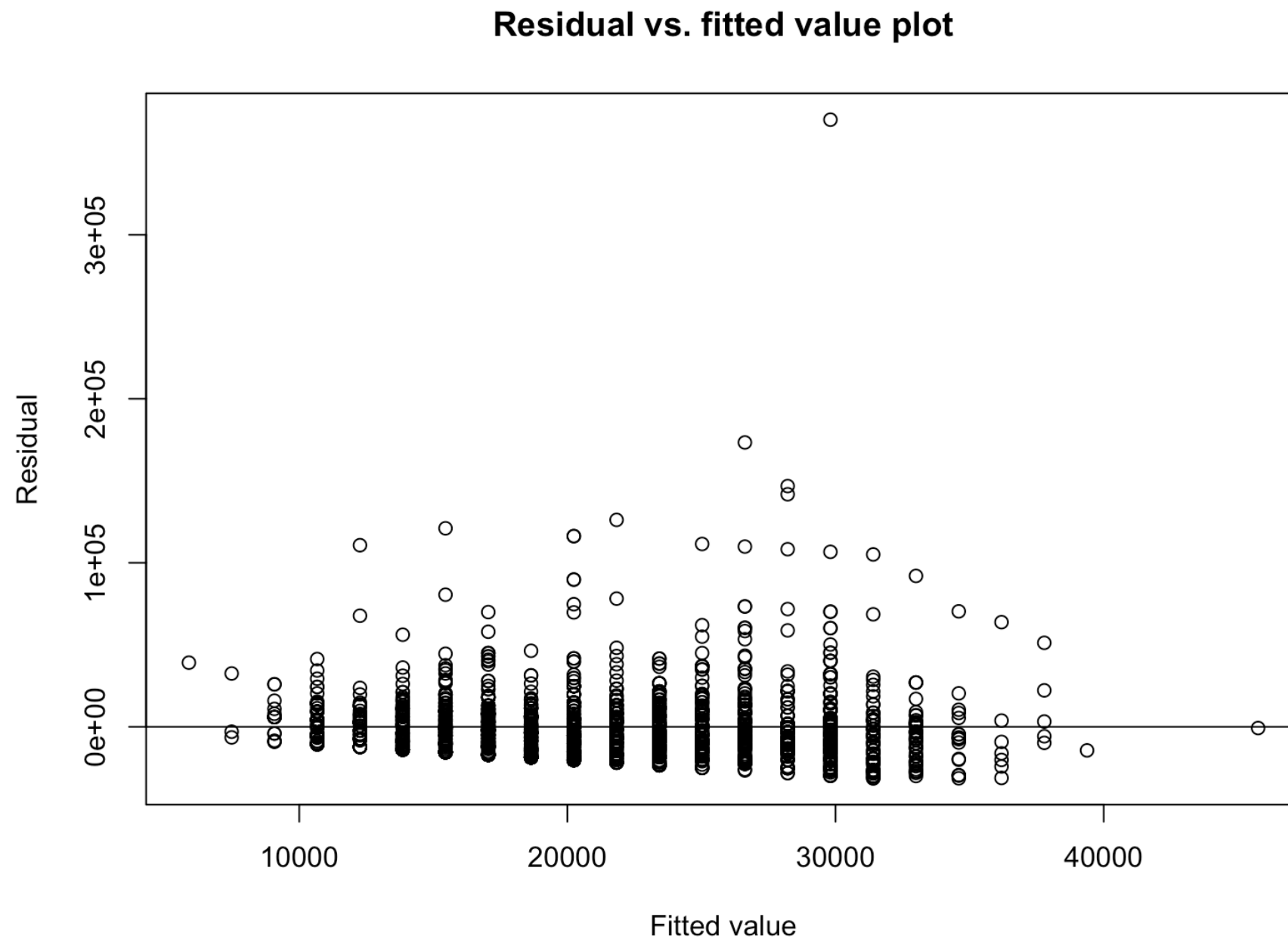
► Code



Assessing model fit: example

Do you notice any issues with the following residual vs. predictor plot?

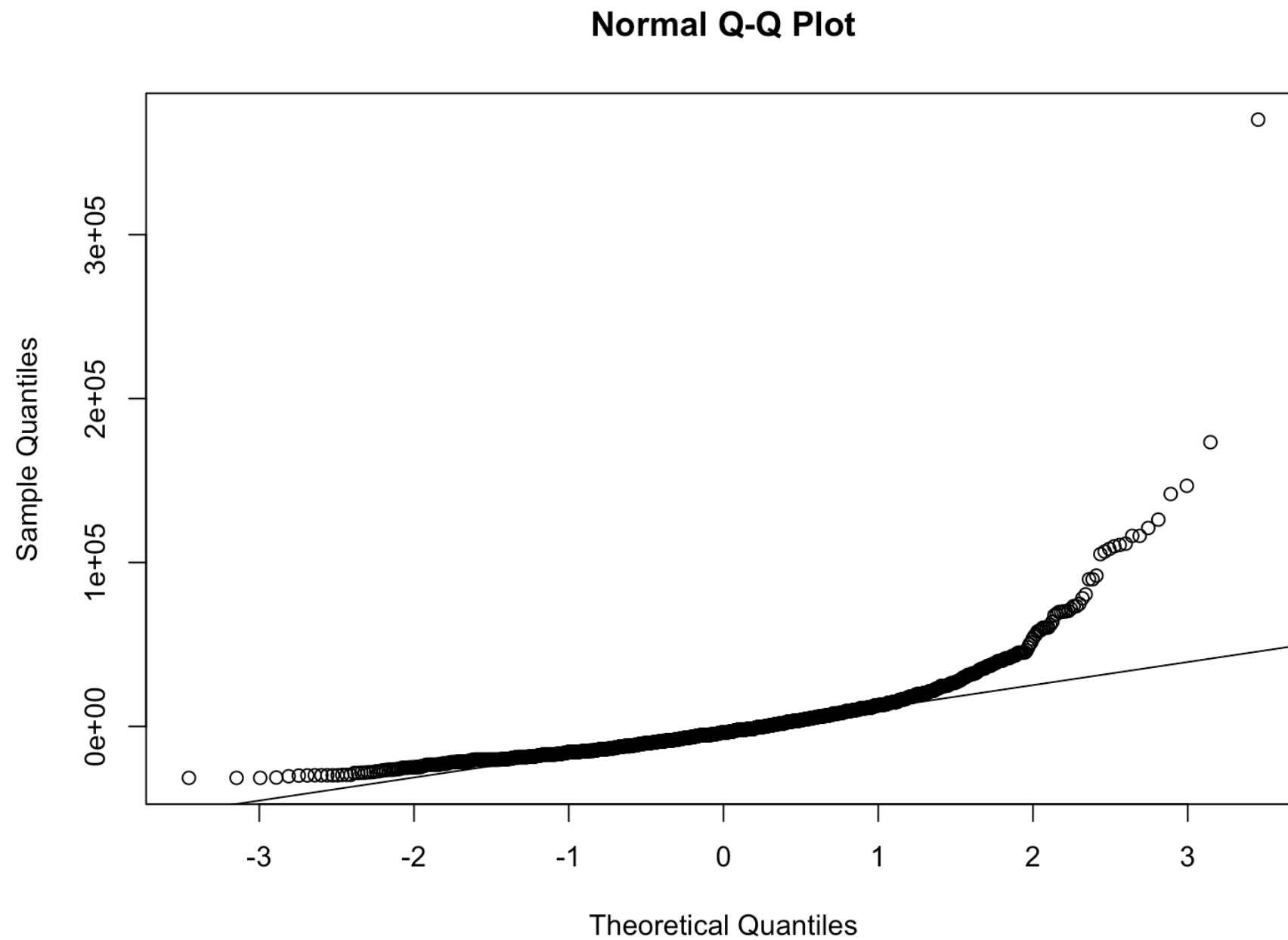
► Code



Assessing model fit: example

Do you notice any issues with the following quantile-quantile plot?

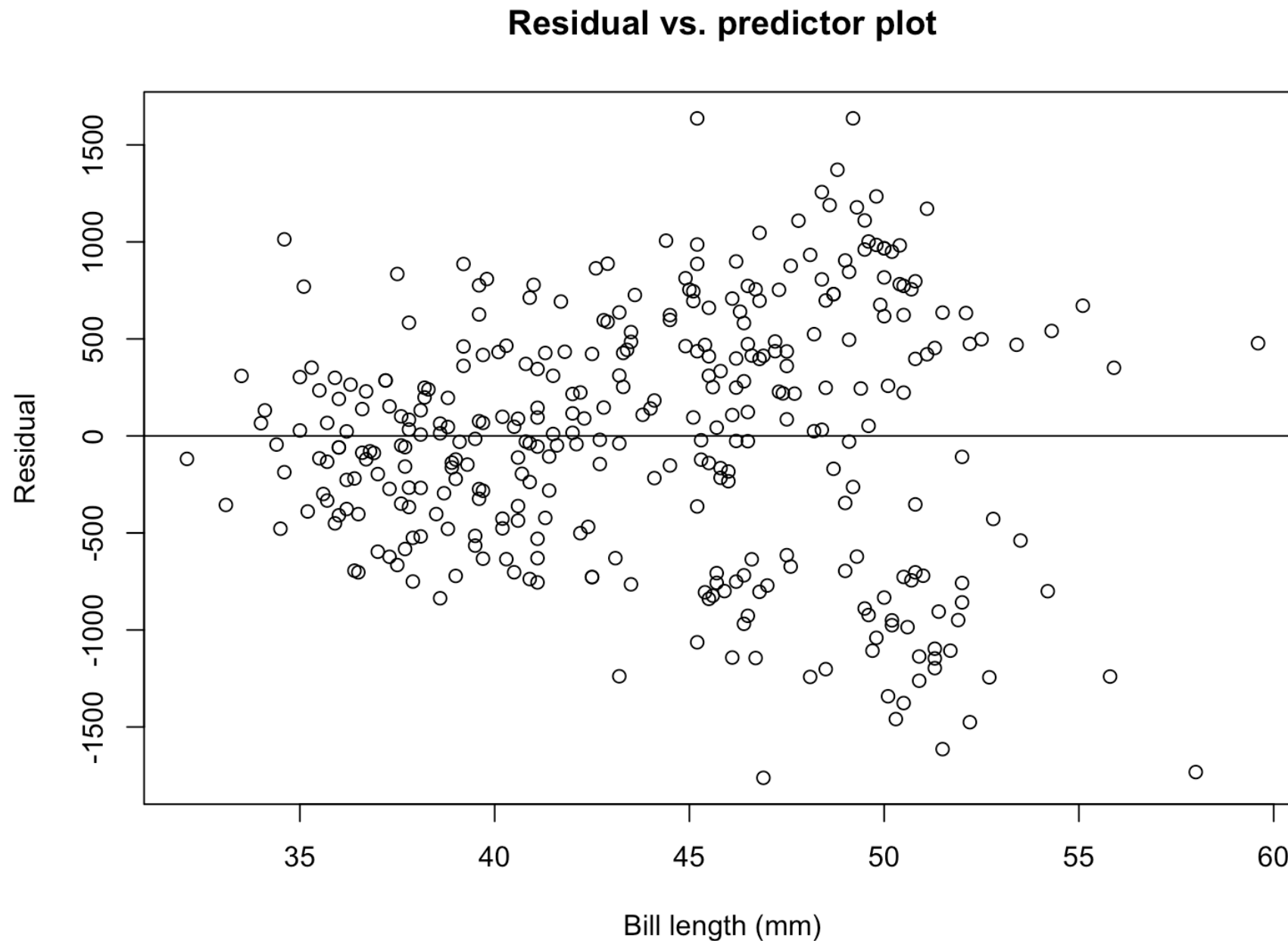
► Code



Assessing model fit: penguins

Do you notice any issues with the following residual plot?

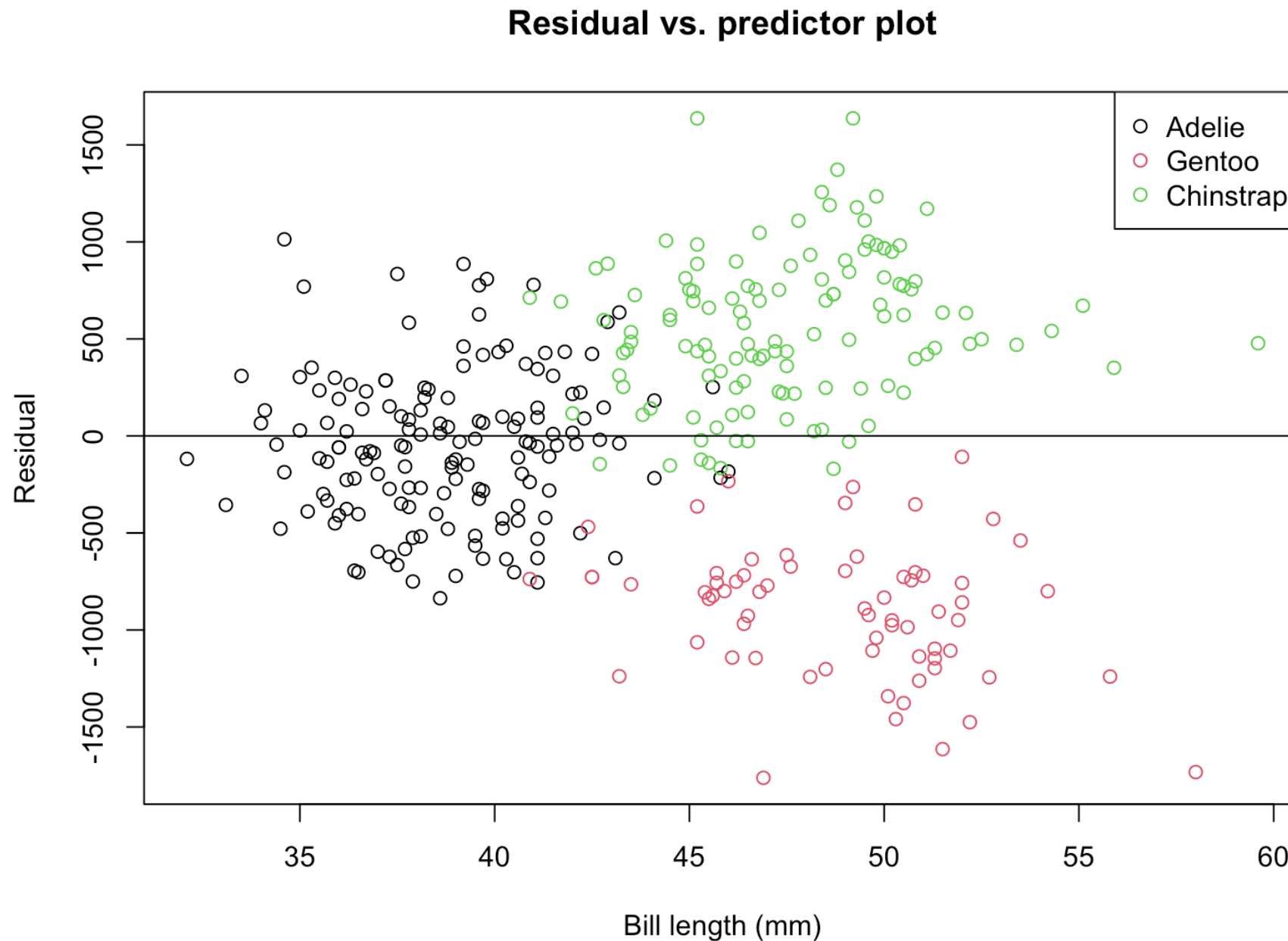
► Code



Assessing model fit: penguins

Sometimes it can help to be creative when assessing model assumptions:

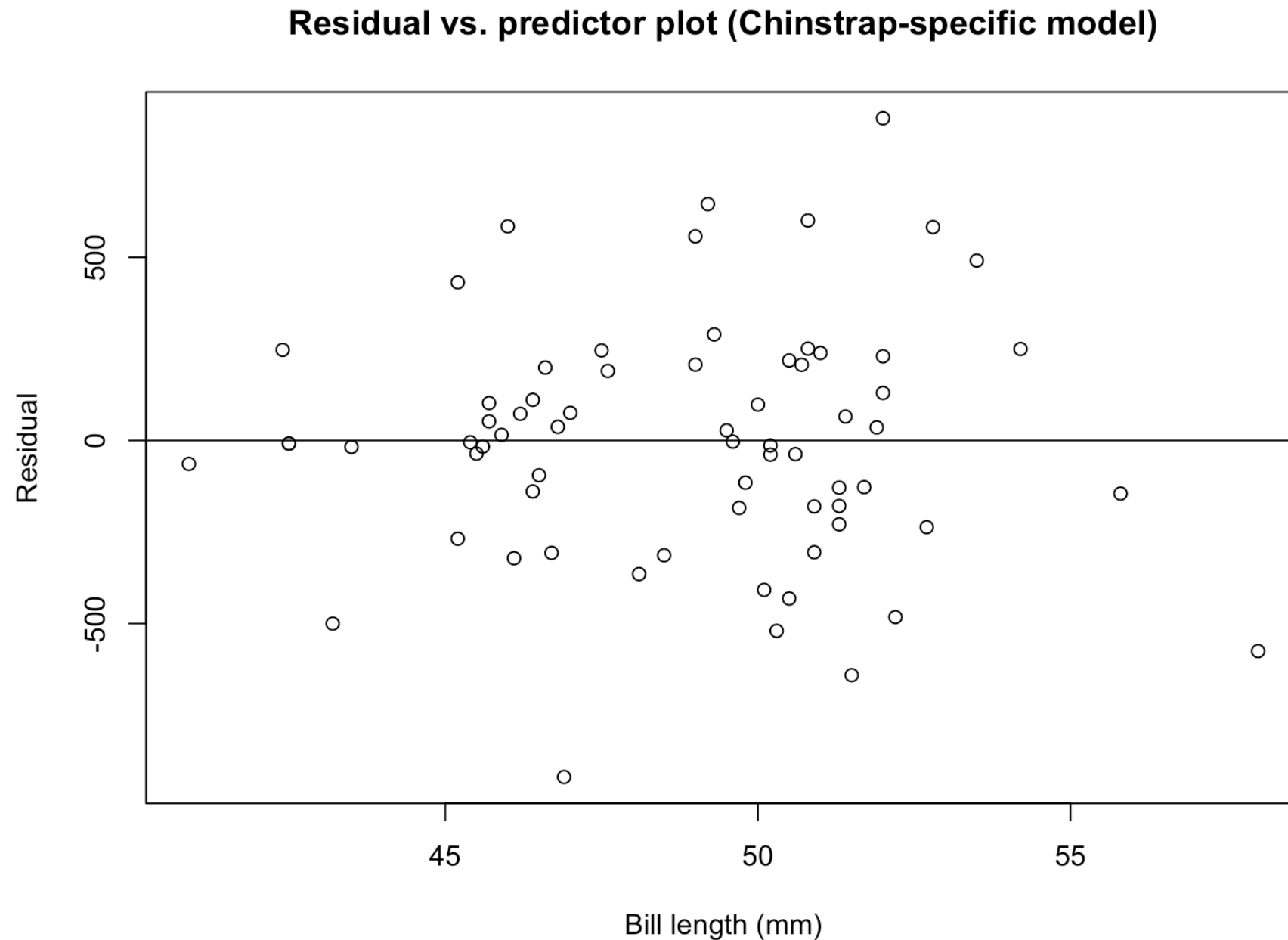
► Code



Assessing model fit: penguins

Sometimes it can help to be creative when assessing model assumptions:

► Code



Assessing model fit using external validation

A direct way to assess model fit is to use it to make predictions and then compare those predictions to actual data. However, to avoid overfitting, we must validate the model using data that are independent and external to the training data set.

If we have no external data, cross validation can also be used to assess the model fit.

Remedial measures

If the simple linear regression model is not appropriate, the two paths forward are:

1. Develop and use a more appropriate model.
2. Transform the data so that the simple linear regression model is appropriate for the transformed data.

KNNL Chapter 3.8 provides an overview of some strategies for dealing with common issues. For now, it is enough to identify the issues and discuss how they may affect your analysis. In later weeks, we will discuss some of these strategies in detail.

Coding exercises

Generate a dataset that breaks one of the assumptions (just one!) of the normal error simple linear regression model. Without revealing the code used to generate the data, swap datasets with a classmate and challenge them to figure out which assumption you broke.

P-Hacking Challenge

Download the data from the pre-course survey on Canvas.

1. Choose a response variable and a predictor for which you believe a linear association exists. Fit a simple linear regression. What are your estimated slope and intercept?
2. Identify the pair of variables which appear to have the strongest linear association. Without terms we haven't covered yet, like "p-value" or "r-squared," explain how you determine what counts as a "strong" linear relationship.